



DEPARTMENT OF ENERGY



BAGONG PILIPINAS

PHILIPPINE ENERGY SITUATIONER AND KEY ENERGY STATISTICS 2024



Philippine Energy Situationer 2024

This issue presents an analysis of energy supply and demand situation in the Philippines for 2024 vis-à-vis 2023. The energy data used herein are based on the Energy Balance Table (EBT) (as of 15 June 2025) as generated by the Policy Formulation and Research Division (PFRD) of the Energy Policy and Planning Bureau (EPPB), unless otherwise stated. Kindly note that *Non-Energy Use* is included in the discussion for Total Final Energy Consumption (TFEC) in this report.



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Abbreviations And Acronyms Used

BBL	Barrels
DOTr	Department of Transportation
GDP	Gross Domestic Product
GHG	Greenhouse gas
GVA	Gross-value added
GWh	Gigawatt-Hour
GWP	Global Warming Potential
ktCO₂e	Thousand tons of carbon dioxide equivalent
kTOE	Thousand tons of oil equivalent
kWh	Kilowatt-hour
LRT	Light Rail Transit
MB	Thousand Barrels
MMB	Million Barrels
MMMT	Million Metric Tons
MMSCF	Million Standard Cubic Feet
MMT	Thousand Metric Tons
MRT	Metro Rail Transit
MTCO₂e	Million tons of carbon dioxide equivalent
MTOE	Million tons of oil equivalent
MW	Megawatt
MWh	Megawatt-hour
ROM	Run of Mine
tCO₂e	Tons of carbon dioxide equivalent
TFEC	Total Final Energy Consumption
TOE	Tons of oil equivalent
TPES	Total Primary Energy Supply
TWh	Terra-watt Hour

I. TOTAL FINAL ENERGY CONSUMPTION (TFEC)

The Philippines ranked among the fastest-growing ASEAN economies in 2024 with a 5.7 percent expansion in Gross Domestic Product (GDP), which was slightly behind Vietnam (7.1 percent) and ahead of major peers such as Indonesia (5.0 percent) and Malaysia (4.4 percent)¹. Economic growth stemmed from increased household spending, continued momentum in the services sector, and favorable investor sentiment.

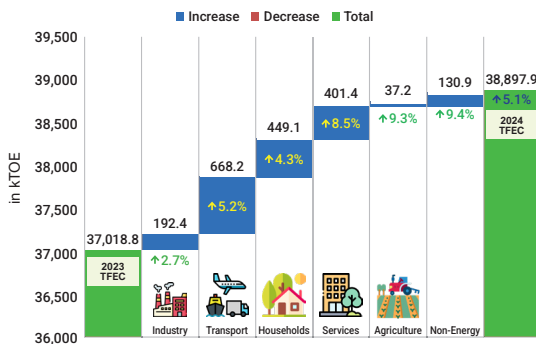


Figure 1: Level (in kTOE) and Percent Changes in TFEC by Sector: 2023 vs 2024

Energy fueled the robust economic activity, as the total final energy consumption (TFEC) rose by 5.1 percent from 37.0 million tons of oil equivalent (MTOE) in 2023 to 38.9 MTOE in 2024 (**Figure 1**). Characterized by increasing consumption across all sectors during the period, with the fastest growths recorded in services and agriculture, forestry, and fisheries (AFF). The aggregate energy use of service establishments rose by 8.5 percent to 5.1 MTOE in 2024 from the previous year's 4.7 MTOE driven by the expansion of retail, finance, and the business process outsourcing (BPO). Despite a decline in overall agricultural output, energy demand went up by 9.3 percent from last year's 401.6 kTOE due to greater reliance on mechanized farming and irrigation systems to support production amid weather-related challenges.

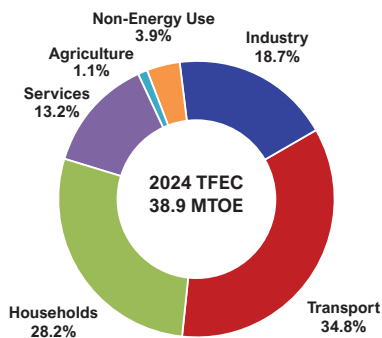


Figure 2: TFEC Shares by Sector, 2024

The transport and household sectors remained the largest energy consumers in absolute terms in 2024, accounting for 34.8 percent and 28.2 percent of total final energy consumption (TFEC), respectively, and registered increased energy use. Transport grew by 5.2 percent, driven by rising vehicle ownership, expanded mobility services, and higher freight demand. Meanwhile, greater availability of modern appliances and other equipment, expanded access to electricity, and higher household income, triggered the 4.3 percent hike in energy consumption in the residential sector (**Figure 2**).

Industrial energy use grew modestly by 2.7 percent, reaching 7.3 MTOE vis-à-vis 7.1 MTOE in 2023. Despite growth in manufacturing and revival of construction activities in 2024, the adoption of energy-efficient technologies and partial automation of

¹ Philippine Statistics Authority. "GDP Expands by 5.2 Percent in the Fourth Quarter of 2024: Brings the Full-Year 2024 GDP Year-on-Year Growth Rate to 5.6 Percent". January 30, 2025. This was later revised to 5.7 percent. For comparison, Vietnam's GDP growth (7.1%) and Malaysia's (5.1%) are cited from General Statistics Office of Vietnam and Bank Negara Malaysia, respectively.

industrial processes tempered the pace of energy consumption in the sector. Meanwhile, utilization of energy commodities as raw materials or feedstocks and inputs to various production process reached 1.5 MTOE, up by 9.4 percent from its year-ago level of 1.4 MTOE.

A. Total Final Energy Consumption by Fuel

Oil and petroleum products continued to dominate the TFEC with an aggregate share of 50.8 percent, as their utilization increased by 4.9 percent from its year-ago level of 18.8 MTOE to 19.8 MTOE (**Figure 3**). The uptrend reflected the country’s reliance on petroleum products, especially for transport which accounted for 65.5 percent share of the total oil demand in 2024, up by 5.1 percent from its 2023 level of 12.3 MTOE. The Services sector notably augmented its consumption of oil and petroleum products with an 11.3 percent of increase in 2024 and 11.9 percent share to the oil demand mix.

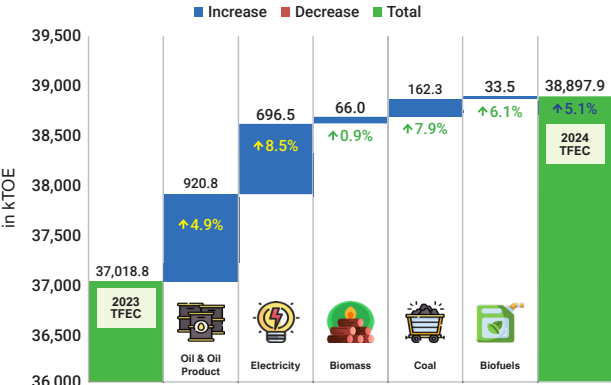


Figure 3: Level (in kTOE) and Percent Changes in TFEC by Fuel 2023 vs 2024

shares, respectively, attributed to steady demand from manufacturing, businesses, and public institutions. Although the AFF and transport sectors accounted for only 3.2 percent and 0.2 percent of total electricity consumption, respectively, both recorded remarkable growth in 2024. Electricity use in AFF surged by 13.2 percent attributed to the increased adoption of electricity-intensive farming technologies, while transport registered the fastest expansion, as electricity consumption soared by 60.4 percent, mainly driven by the growing adoption of electric vehicles.

On the other hand, consumption of biomass such as fuelwood, charcoal, and agricultural residues stood at 7.3 MTOE in 2024, which represented 18.9 percent of the energy demand mix during the year (**Figure 4**). Despite its continued use among low-income and off-grid households for traditional cooking, overall biomass consumption grew modestly by 0.9 percent. Establishments and industrial facilities made up the remaining share of biomass utilization, but their growth remained modest at only 0.7 percent and 0.3 percent, respectively.

Coal use for industrial processes and production inputs contributed 5.7 percent share to TFEC as its levels reached 2.2 MTOE in 2024, or 7.9 percent higher than its 2.1 MTOE in

Electricity consumption registered an 8.5 percent increase in 2024, reaching 8.9 MTOE and comprising 23.0 percent of TFEC (**Figure 4**). Households remained as the largest electricity user with a share of nearly 40 percent to the total electricity consumption (TEC). The Industry and Services sectors contributed 30.0 percent and 27.0 percent

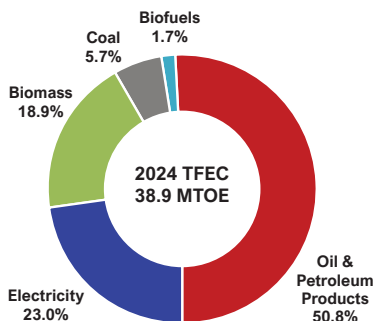


Figure 4: TREC Shares by Fuel, 2024

2023. Industries engaged in non-metallic mineral processing, cement, and other heavy industries accounted for the bulk of coal demand.

Parallel with the increased utilization of gasoline and diesel, consumption of biodiesel reached 182.6 kTOE, while bioethanol totaled 461.8 kTOE, as both fuels made vital contributions in decarbonizing the transport sector during the year. Collectively, bioethanol and biodiesel accounted for 1.7 percent of TREC in 2024.

B. Total Final Energy Consumption by Sector

1. Transport

Energy consumption of the transport sector sustained its upward trajectory with a 5.2 percent increase to 13.5 MTOE in 2024 from 12.9 MTOE in 2023. This was attributed to several key factors such as expanded infrastructure, increased vehicle ownership, and rising freight activity. Specifically, the surge in electric vehicle use, supported by zero-tariff policies, also contributed to higher electricity demand and gradual adoption of cleaner technologies.

Road transport's energy consumption grew by 6.0 percent to 12.3 MTOE, with a 90.7 percent share to the sector's energy mix (**Figure 5**). The uptrend compensated for the decline in energy consumption across other transport subsectors and was supported by robust growth in vehicle sales, which increased by 8.7 percent in 2024, according to a joint report by the Chamber of Automotive Manufacturers of the Philippines, Inc. (CAMPI) and the Truck Manufacturers Association (TMA)². The rise in new vehicle registrations—which spanned private cars, commercial vehicles, and delivery fleets—reflected growing mobility demand, urbanization, and increased economic activity.

Rail transport reported a 6.2 percent drop in energy use, from 11.9 kTOE in 2023 to 11.2 in 2024, due to a double-digit reduction in both diesel and biodiesel of more than 40 percent attributed to the suspension of Philippine National Railways (PNR) operations in Metro Manila³, which took effect on 28 March 2024. The five-year service halt is intended to expedite the construction of the North-South Commuter Railway (NSCR). On the other hand, the surge in passenger ridership for both Metro Rail Transit (MRT) and Light Rail Transit Authority (LRTA) systems – which reached over 189 million passengers in 2024 from 170 million in 2023^{4,5} slightly cushioned the downtrend as electricity use for rail improved by 3.4 percent.

² JON VIKTOR D. CABUENAS, GMA Integrated News & JON VIKTOR D. CABUENAS, GMA Integrated News. (2025b, January 14). Over 467,000 vehicles added to Philippine streets in 2024. GMA News Online. https://www.gmanetwork.com/news/money/motoring/932813/over-467-000-vehicles-added-to-philippine-streets-in-2024/story/?fbclid=IwY2xjawLUda1leHRuA2FibQlxAiBicmlkETFORGY4cU5wcjduMzRcWhiAR4bxTn2Zod721pJo6ETKh9_3UfY-I7C6DhRq0orwz-x24ioPs0HsYgpbWx2Zg_aem_zn4QCUCDehabyDGslnkH9Q

³ Yu, L. S. (2024, March 8). PNR will stop operations on March 28 for five years. RAPPLER. <https://www.rappler.com/business/pnr-stop-operations-march-28-2024/>

⁴ DOTR-MRT3. (n.d.-b). <https://dotmrt3.gov.ph/news/mrt-3-logs-135-million-riders-in-2024>

⁵ LRTA ACCOMPLISHMENT REPORT CY 2024. (n.d.). In <https://www.lrt.gov.ph/wp-content/uploads/2025/04/Annual-Accomplishment-Report-CY-2024-04.02.25.pdf>



Energy consumption in domestic water and air transport declined by 0.7 percent and 6.0 percent, respectively, due to disruptions in trade and logistics operations. These were primarily caused by the suspension of various shipping and airline routes during the series of typhoons and storms in the latter part of 2024⁶. In the aviation subsector alone, at least 184 flights were cancelled during major weather events such as Super Typhoon Carina and Tropical Storm Trami, based on official advisories from the Manila International Airport Authority (MIAA)⁷.

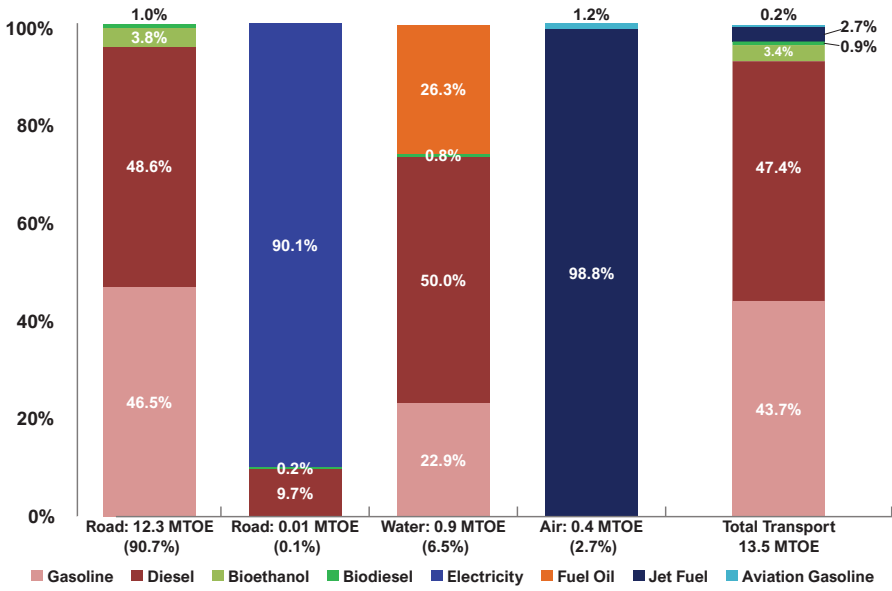


Figure 5: Transport FEC: by Sector and Fuel Shares, 2024

Petroleum-based fuels remained the primary energy source for the transport sector. Demand for gasoline and diesel, with a collective share of 91.1 percent of transport energy consumption, went up by 5.9 percent and 5.1 percent, respectively. This is attributed to increased road traffic, higher vehicle sales—particularly of fuel-powered cars and delivery trucks—and the continued reliance on diesel for freight and public transport operations. Meanwhile, fuel oil use, specifically for domestic maritime activities, went up by 5.5 percent in 2024 as operators favored larger, cost-efficient vessels amid weather disruptions. Consumption of aviation fuels dropped by 6.0 percent due to flight cancellations from severe weather and airline fuel-saving measures.

Consistent with the implementation of cleaner energy policies, such as biofuel blending and electrification of the transport sector, the country continued its steady shift to alternative energy sources for transport in 2024. Electricity consumption rose to 60.4 percent, primarily due to the expanded use of EVs, with over 18,690 additional units

6 Desiderio, L. (2025, March 22). Domestic trade value down in Q4. Philstar.com. <https://www.philstar.com/business/2025/03/23/2430327/domestic-trade-value-down-q4>

7 Reuters. (2024, October 25). Tropical Storm Trami kills 46 in Philippines, could return next week. NST Online. <https://www.nst.com.my/world/region/2024/10/1125327/tropical-storm-trami-kills-46-philippines-could-return-next-week>

8 Zurbano, J. E. (2024, July 25). MIAA: 148 flights cancelled in Manila due to Super Typhoon 'Carina'. Manila Standard. <https://manilastandard.net/news/314475985/miaa-148-flights-cancelled-in-manila-due-to-super-typhoon-carina.html>



sold in 2024⁹. Compliance with the new mandated 3 percent blend that took effect in October 2024, biodiesel demand levels improved by 5.1 percent to 126.8 kTOE from last year’s 120.6 kTOE. Consumption of bioethanol reached 461.8 kTOE in 2024, 5.6 percent more than its 2023 level of 437.1 kTOE. These uptrend remained consistent with national goals of reduced dependence on imported fuels and promotion of alternative fuels for transportation.

2. Households

In 2024, the household sector maintained its position as the second-largest energy-consuming sector in the Philippines with its 28.2 percent share to TFEC. Energy use for household activities increased by 4.3 percent to 11.0 MTOE from 10.5 MTOE in 2023. This upward trend was largely attributed to enhanced access to energy services, driven by the government’s sustained electrification initiatives, expanded availability of clean cooking fuels, and programs promoting energy efficiency at the household level.

Biomass remained the primary fuel source, with fuelwood accounting for 37.8 percent of household energy use (**Figure 6**), followed by charcoal at 13.0 percent and agricultural waste at 4.4 percent. Although total biomass consumption grew slightly by 1.0 percent from 6.0 MTOE in 2023 to 6.1 MTOE in 2024, its share to the household energy mix declined by 1.8 percentage points. This shift reflected a gradual transition toward cleaner and more modern fuels, such as electricity and LPG, driven by improved electrification and wider access to clean cooking technologies.

Successfully electrified over 2,000,000 additional household reported by the Distribution Utilities (DUs) as of the first half of 2024, the Department of Energy (DOE) continued the implementation of its Total Electrification Program¹⁰. This development led to an 11.5 percent increase in household electricity consumption, which reached 3.5 MTOE in 2024 and accounted for 32.3 percent of the household energy mix. The increased adoption of electricity, recognized as a cleaner and more efficient fuel, reflected a growing preference for modern energy sources and marked a shift away from traditional fuels.

Liquefied petroleum gas (LPG) consumption rose modestly by 1.9 percent to 1.3 MTOE in 2024 and comprised 12.2 percent of the household energy demand mix. Meanwhile, kerosene use declined by 6.2 percent to 29.9 kTOE in 2024. These trends reflected the impact of improved electrification and expanded access to cleaner technologies on household fuel choices.

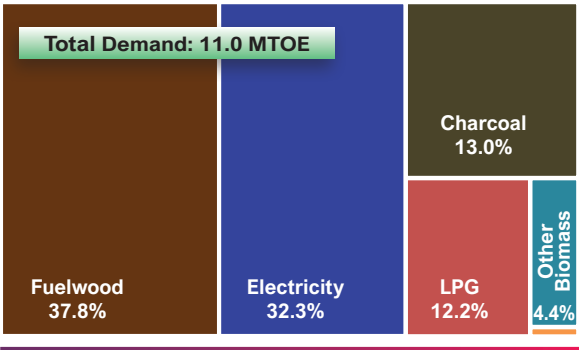


Figure 6: Household Energy Demand Mix, by Fuel Shares: 2024

9 Mioten, E. (2025, March 3). The Philippines’ push for electric mobility. BusinessWorld Online. <https://www.bworldonline.com/special-features/2025/02/28/656679/the-philippines-push-for-electric-mobility/>

10 2024–2033 National Total Electrification Roadmap, DOE

3. Industry

Aggregate industrial output sped up by 5.6 percent in 2024 vis-à-vis 3.6 percent in 2023, and contributed 29.1 percent share to GDP. This acceleration was primarily driven by robust performances in the construction and manufacturing sectors. Correspondingly, the sector’s energy consumption rebounded by 2.7 percent to 7.3 MTOE, reversing the 0.5 percent decline recorded in the previous year.

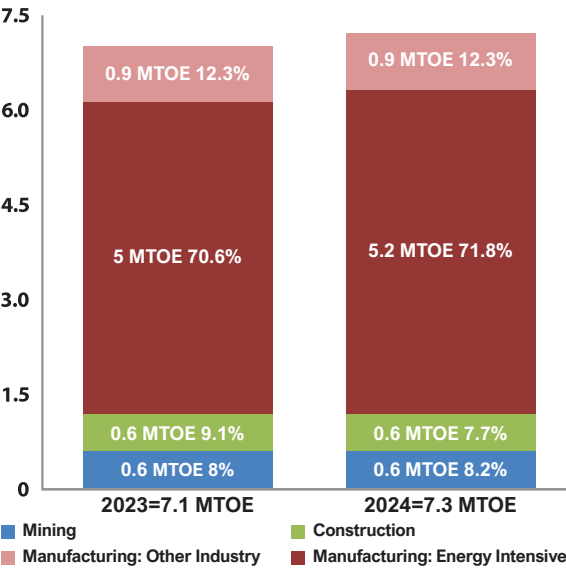


Figure 7: Industry Energy Demand Mix, by Sub-sector: 2023 vs 2024

Energy use in the manufacturing sub-sector, which consistently accounted for more than 80.0 percent share (**Figure 7**) to industry total, rose by 4.2 percent as production scaled up to meet increased demand for food products, consumer goods, and construction materials. Energy-intensive manufacturing sub-sectors recorded an accumulated 5.2 MTOE of energy utilization or 4.5 percent higher than its 2023 level of 5.0 MTOE. This was due to higher energy use reported in cement (30.7 percent), food processing (0.9 percent) and machinery/equipment (1.4 percent).

Despite the 5.3 percent hike in approved permits for residential and commercial buildings¹¹, energy use in construction declined by 12.8 percent due to double-digit reduction in diesel utilization. This implied reduced reliance on diesel-powered equipment, and a shift to electric-powered and fuel-efficient machinery, as well as other alternative construction technologies.

On the other hand, despite the sluggish performance of the mining sub-sector, energy consumption went up by 5.2 percent to 596.3 kTOE in 2024 from its year-ago level of 566.8 kTOE. This is attributed to sustained energy use in extraction and processing requirements to address operational challenges such as adverse weather conditions and logistical issues.

Electricity accounted for the largest share in the sector’s aggregate energy demand mix at 36.8 percent. Its aggregate consumption went up by 5.4 percent to 2.7 MTOE in 2024 (**Figure 8**) from 2.5 MTOE in the previous year due to its extensive use across production activities and processes. Lower coal prices contributed to the 8.0 percent rise in coal consumption during the year, as levels reached 2.1 MTOE from 2.0 MTOE in 2023. Coal contributed 29.6 percent share to industry’s energy demand,

¹¹ Philippine Statistics Authority. (n.d.). Construction statistics from approved building permits [Web page]. Retrieved July 22, 2025 from <https://psa.gov.ph/statistics/construction/pcs>



as it was primarily used to fuel high-temperature processes, such as cement as it was primarily used to fuel high-temperature processes, such as cement manufacturing where it registered a 33.7 percent hike in utilization during the year. Total industrial demand for oil and petroleum products dropped by 6.5 percent to 1.5 MTOE in 2024 from 1.6 MTOE a year-ago. The downtrend is attributed to contractions recorded in the use of diesel (8.2 percent), fuel oil (0.3 percent), LPG (5.6 percent) and kerosene (20.0 percent) as industries adjusted their operating expenses due to rising fuel prices. Biomass remained as an important energy commodity with 12.8 percent share of total energy demand, albeit a minimal 0.3 percent increment in its consumption for 2024. Consumption of biodiesel remained on the downtrend and dropped further to 19.5 kTOE in 2024 from 21.4 kTOE in 2023.

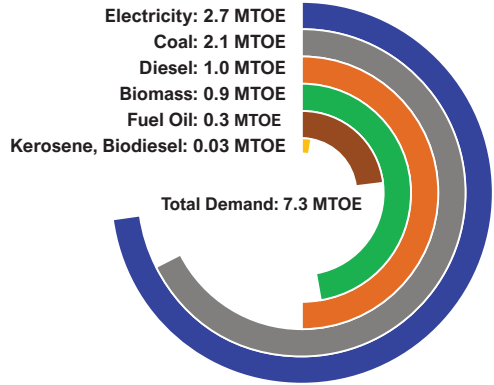


Figure 8: Industry Energy Demand Mix, by Fuel: 2024

4. Services¹²

The Services sector registered a 6.7 percent growth in aggregate value-added and outpaced both the industry and agriculture sectors in 2024. It was a major driver of economic growth with its significant 62.9 percent share to the overall economy. Domestic trade, particularly wholesale and retail¹³, was buoyed by strong consumer demand and increased adoption of e-commerce and digital technologies. Thus, the sector's posted the second-fastest acceleration vis-à-vis other end use sectors at 8.5 percent to 5.1 MTOE from its 2023 level of 4.7 MTOE.

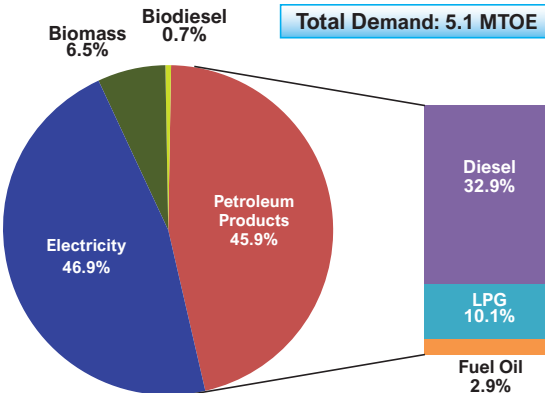


Figure 9: Services Energy Demand Mix, by Fuel Shares: 2024

Electricity demand reached 2.4 MTOE or 6.9 percent higher than its year-ago level of 2.3 MTOE, and accounted for close to half (46.9 percent) of the sector's energy demand mix for 2024. It signaled the continued reliance of the sector on electricity as fuel for lighting and space cooling requirements in offices, retail shops, hotels, and healthcare facilities. Diesel, used extensively as fuel for backup generators to ensure operational continuity,

¹² Excludes Transport and Storage

¹³ Isip, I. (2025, January 13). Retailers, wholesalers see 1-10 % growth this yr – survey. Malaya Business Insight. Retrieved October 10, 2025, from <https://malaya.com.ph/business/business-news/retailers-wholesalers-see-1-10-growth-this-yr-survey/>

emerged as the most widely used oil product in the services sector in 2024 and comprised 32.9 percent of the sector’s energy consumption. Liquefied petroleum gas (LPG) registered a 6.7 percent increase in usage that reached 519.4 kTOE, comprised the 10.1 percent of the sector’s energy demand. This was largely due to sustained use in restaurants, catering services, and other commercial establishments for cooking fuel. In contrast, fuel oil utilization fell by 2.4 percent attributed to the transition toward more cost-effective or environmentally friendly options. Meanwhile, biomass use in the sector inched up by 0.7 percent, as it maintained its utilization among food establishments.

5. Agriculture

Energy consumption in the agriculture sector reached 438.8 kTOE in 2024 or 9.3 percent higher than its 401.6 kTOE in 2023, reflected a share of 1.1 percent of the country’s TREC. The hike in the sector’s energy use mirrored the shift to mechanized operations, expanded livestock farming, as well as resilient harvests that required processing and transport and increased use of cold storage and processing facilities.

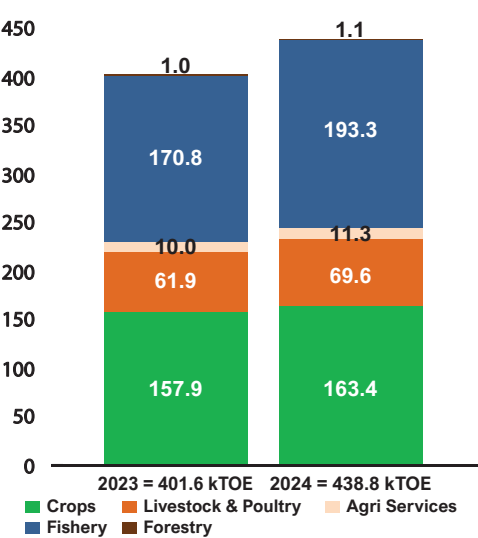


Figure 10: AFF Energy Demand Mix, by Sub-sector (kTOE): 2024

The agri-industry subsector’s energy demand went up by 6.3 percent to 244.3 kTOE compared to its year-ago level of 229.8 kTOE. This growth primarily stemmed from heightened activity in crop production and animal raising¹⁴. The combined shares of crops, livestock and poultry and agri services totaled more than half (55.7 percent) of the agriculture sector’s energy requirement (Figure 10). On the other hand, energy use in livestock and poultry rose by 12.5 percent to 69.6 kTOE in 2024, driven mainly by commercial poultry farms that relied mainly on electricity for feed processing, water, and ventilation. Agricultural services also saw a 13.2 percent rise in electricity use for post-harvest and logistics operations.

Energy consumption in the fishery sub-sector went up by 13.2 percent to 193.3 kTOE in 2024. The uptake was due to increased reliance on cold storage and aeration systems, as well as expanded fishing activities in areas recovering from the past climate disruptions. Meanwhile, forestry, though a minor component of agricultural energy use, increased slightly from 1.0 kTOE to 1.1 kTOE, consistent with small gains in logging and wood processing activities in upland areas.

Electricity accounted for more than two-thirds (65.6 percent) of the agriculture sector’s energy demand mix and saw a significant surge of 13.2 percent in 2024 (Figure 11). It highlighted the growing reliance on electricity for modern farming

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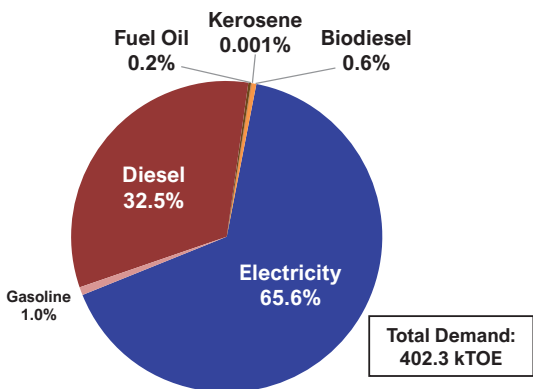


Figure 11: AFF Energy Demand Mix, by Fuel Shares: 2024

practices like irrigation and cold storage. On the other hand, diesel remained the most consumed liquid fuel with 32.5 percent share and posted a 7.0 percent increase given its role as fuel for agricultural machinery. Combined consumption of gasoline, fuel oil and kerosene, with a minimal share of 1.2 percent, dropped by more than twice its 2023 level to 5.3 kTOE due to increased preference for more efficient and clean fuels such as electricity.

II. TRANSFORMATION

A. Oil Refining

The Philippine oil refining sector remained dependent on a single operational facility—Petron’s Bataan Refinery, albeit 6.5 percent decline in refinery output from 6.5 MTOE (47.7 MMB) in 2023 to 6.0 MTOE (50.9 MMB). This reduction stemmed from operational challenges and strategic adjustments, as well as scheduled maintenance shutdowns that affected the plant’s refinery operations. In addition, the hike in importation of finished petroleum products effectively reduced the immediate need for higher domestic refining output.

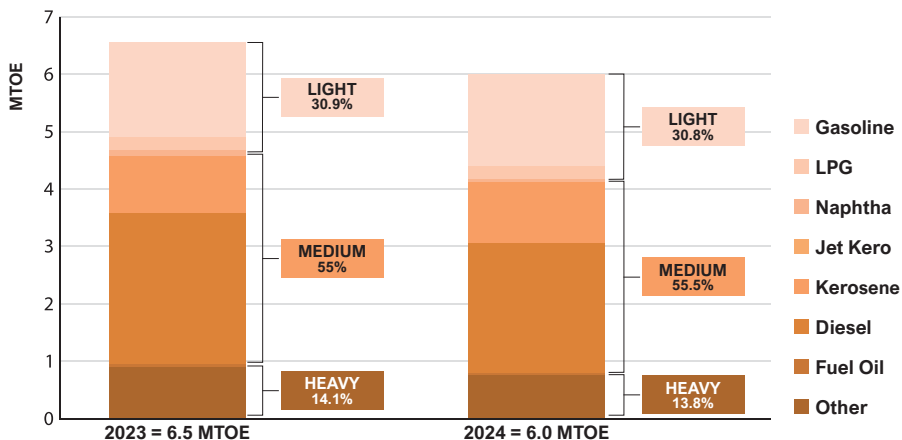


Figure 12. Refinery Production, by Fuel (in MTOE), 2023 vs. 2024

The volume of domestically refined diesel and gasoline, with an aggregate share of 64.1 percent to the country’s refinery mix, contracted by 12.5 percent and 0.9 percent, respectively, as imports met a larger share of market demand. LPG (3.8 percent share), kerosene (0.3 percent share) and naphtha (0.3 percent share) exhibited the same downtrend with negative growths of 2.4 percent, 7.2 percent and 87.9 percent, respectively. Aggregate output of other petroleum products, which accounted for 12.9 percent share, also fell by 10.4 percent. In contrast, refinery output for jet kerosene (0.3. percent share) improved by 12.8 percent, while that of fuel oil (0.9 percent share) increased by 20.7 percent.

B. Power Generation and Fuel Input

Aggregate power generation reached 126.9 terawatt-hours (TWh) in 2024, 7.6 percent more than its 2023 level of 118.0 TWh. The increase was in response to robust economic performance and continued expansion in electricity demand across sectors during the year.

Coal-fired power plants remained the dominant source of supply, accounting for 62.5 percent (**Figure 13**) of the total generation mix or 79.4 TWh in 2024, a significant increase of 7.6 percent from its 2023 level of 73.8 TWh. Natural gas power generation also grew steadily, contributing 18.0 TWh or 14.2 percent of total generation, up from its previous year’s level of 16.7 TWh. Oil-based power plants also posted increased in their generation output at 2.9 percent.

Meanwhile, renewable energy sources supplied 22.2 percent of the total generation output in 2024. Geothermal and hydropower, with combined shares of 17.1 percent, each contributed more than 10 TWh. Solar energy showed a notable 49.8 percent growth, as it produced 3.8 TWh of power. Wind power dropped by 5.3 percent to 1.2 TWh with only 1.0 percent share, while biomass generation stood at 1.4 TWh, equivalent to a 1.1 percent share of the generation mix.

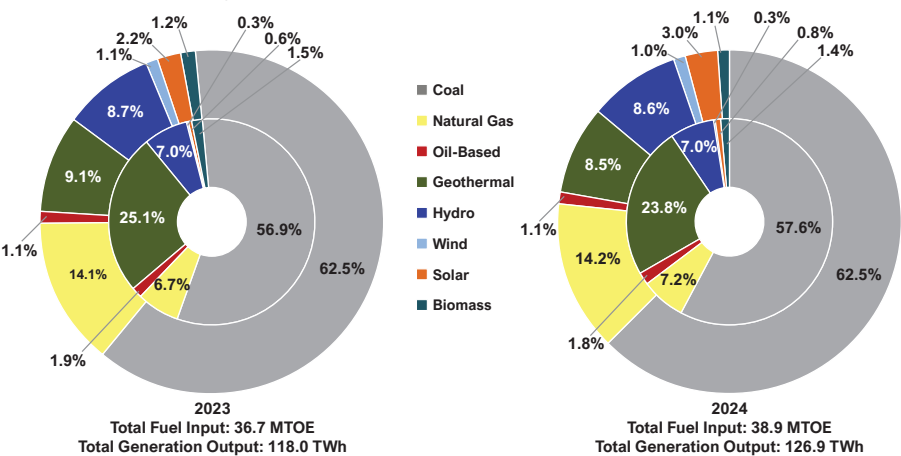


Figure 13. Power Generation Output and Fuel Input, by Fuel Shares: 2023 vs 2024



Power plants' total fuel input rose by 6.0 percent to 38.9 MTOE in 2024, from 36.7 MTOE in 2023. Due to higher demand from baseload operations, coal consumption for power generation went up by 7.3 percent to 22.4 MTOE and accounted for 57.6 percent of the total fuel input (**Figure 13**). Natural gas increased by 13.9 percent to 2.8 MTOE from 2.5 MTOE in 2023 spurred by LNG imports and additional gas-fired capacity that came online during the year. Geothermal energy input remained relatively stable at 9.3 MTOE, while hydro increased slightly by 6.0 percent to 2.7 MTOE during the period. Solar and wind inputs for power generation mirrored their generation output growth trend, with inputs of 0.3 MTOE and 0.1 MTOE, respectively. Meanwhile, biomass consumption for power applications rose modestly by 2.5 percent from 0.5 MTOE in 2023 to 0.6 MTOE in 2024.

III. TOTAL PRIMARY ENERGY SUPPLY (TPES)

In 2024, TPES rose to 69.0 MTOE, a 5.2 percent increase from 65.6 MTOE in 2023, driven by stronger energy demand and expanded power generation. Although the growth in net energy imports slowed to 8.2 percent in 2024 vis-à-vis 13.9 in 2023, it remained substantial with 55.0 percent share to TPES at 38.0 MTOE. It compensated for the sluggish 1.7 percent increment in domestic energy production as it reached 31.0 MTOE in 2024 from 30.5 MTOE the previous year. As a result, energy self-sufficiency declined further by 1.6 percentage points to 45.0 percent from 46.5 percent in 2023 (**Figure 14**).

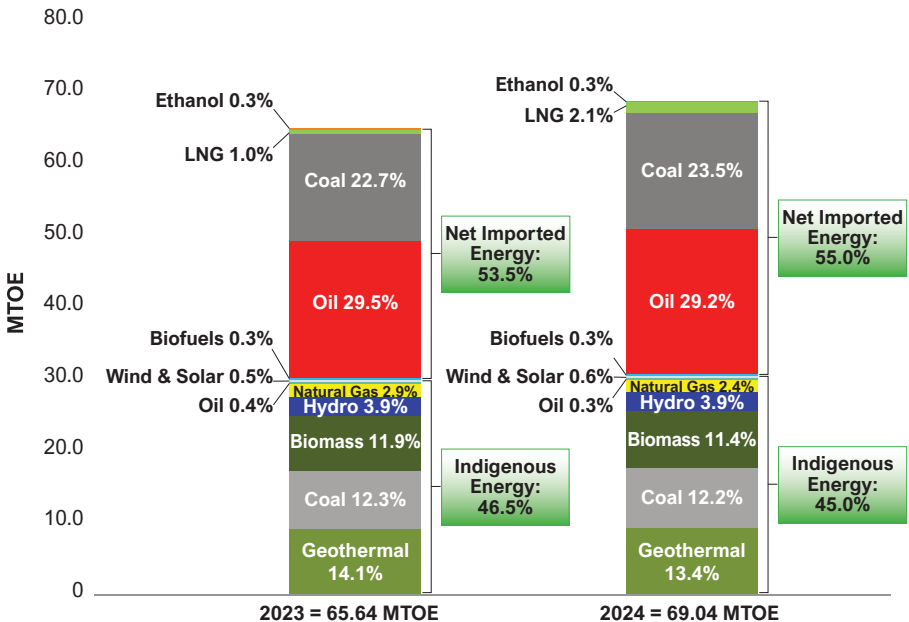


Figure 14. TPES, by Fuel Shares: 2023 vs 2024

Coal remained the dominant energy source with 35.7 percent share to TPES or 24.6 MTOE, which was 7.3 percent more than its year-ago level of 23.0 MTOE. The volume of local and imported coal ramped up by 4.3 percent and 8.9 percent, respectively, to provide for the demand from power and industrial processes. Aggregate renewable energy sources (geothermal, hydro, solar, wind, biomass, and biofuels) posted a 2.1 percent expansion as it reached 21.0 MTOE or 30.4 percent share to the country's energy mix. Meanwhile, total oil supply rose by 3.9 percent, from 19.6 MTOE in 2023 to 20.4 MTOE in 2024, and contributed 29.5 percent to TPES. Bulk of the increase is attributed to the continued reliance on imported petroleum products, as domestic production remained limited. Meanwhile, natural gas supply significantly increased by 21.0 percent boosted by the double-digit hike in LNG imports during the year.

A. Indigenous Energy

The uptrend in domestic coal production and renewable energy sources, such as hydro, solar and biofuels, offset the decline in oil, natural gas and wind to keep the level of total indigenous energy production afloat at 31.0 MTOE or 1.7 percent higher than 30.5 MTOE recorded in 2023. This was an improvement to the slowdown exhibited in 2023, and reflected the country's continued efforts to strengthen energy security through aggressive promotion indigenous energy sources.

Fossil Fuels

i. Oil

Combined production of crude oil and condensate fell by 9.4 percent to 241.2 kTOE (2.2 million barrels (MMB)) in 2024 from 266.2 kTOE (2.4 MMB) in 2023. Galoc oil field, the country's only commercially producing oil field during the year, recorded a 10.6 percent reduction in crude oil production due to aging infrastructure and the progressive depletion of its reservoir consistent with its late-life production phase. Condensate production from the Malampaya gas field, which is fully exported, also dropped by 9.0 percent, consistent with the field's ongoing decline in natural gas output and associated liquids.

ii. Coal

Coal production increased by 4.3 percent in 2024 to 8.4 MTOE (15.9 million metric tons (MMMT)) from 8.1 MTOE (15.3 MMT) in the previous year. Semirara Mining and Power Corporation (SMPC), the country's largest coal producer, sustained its production level at 14.7 MMT and accounted for 93.7 percent of total domestic production. Meanwhile, coal mines in South Cotabato, Sarrangi and Sultan Kudarat and Zamboanga reported a surge of as much as four (4) times their combined production level in 2023 to 0.9 MMT and translated to a share of 5.5 percent to total domestic coal production. Small-scale mines and those in Cebu, Negros and Bicol contributed an additional 0.1 MMT or 0.8 percent share.



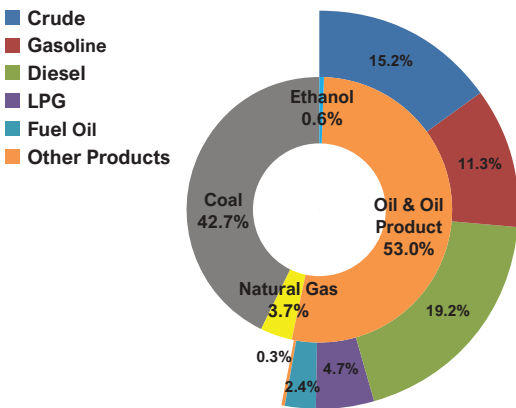
v. Wind

Wind energy fell by 5.3 percent to 106.5 kTOE in 2024 compared to 112.5 kTOE in 2023. While it accounted for 0.2 percent of the energy mix during the year, offshore wind (OSW) development remained on the upswing. As of December 2024, 291 OSW service contracts have been issued which are expected to expand the country’s current installed capacity of 426.9 MW²⁰.

vi. Biofuels

In 2024, local production of biofuels continued to rise in response to increased demand for transport fuels and compliance with Biofuels Act of 2006. Biodiesel production reached 218.8 kTOE or 17.6 percent higher than its year-ago level, while bioethanol output increased by 1.2 percent to 216.6 kTOE. These expansions reflected the operational performance of twelve (12) biodiesel production facilities with a combined annual capacity of 615.3 million liters per year (MLPY)²¹, as well as twelve (12) bioethanol plants, with an aggregate capacity of 438.0 MLPY²².

B. Net Energy Imports



The volume of total net energy inflows rose by 8.6 percent to 38.0 MTOE in 2024, up from 35.1 MTOE in the previous year. This surge helped sustain the country’s energy needs amidst rising domestic demand and subdued local energy production during the year. Crude oil and finished petroleum products accounted for the largest share at 53.0 percent, followed closely by coal with 42.7 percent share, while liquefied natural gas (LNG) and ethanol contributed the remaining portion (4.3 percent) (Figure 15).

Figure 15. Net energy imports, by Fuel Shares: 2024

Combined import volume of crude and finished petroleum products in 2024 slowed down by 2.1 percent compared to the 4.9 percent increase registered in 2023. Total oil imports reached 22.9 MTOE, as finished petroleum products accounted for 73.4 percent and crude oil contributed the remaining 26.6 percent. Crude oil imports fell by 4.5 percent from 6.4 MTOE in 2023 to 6.1 MTOE during the year, attributed to constrained domestic refining, geopolitical tensions and regional economic uncertainties. This contraction was offset by the 4.6 percent hike imports of finished petroleum products, which emerged as a more economically viable solution, and helped sustain requirement of the transport sector, specifically for diesel and gasoline.

20 Ibid

21 Department of Energy (Philippines), Renewable Energy Management Bureau, Biomass Energy Management Division. (2024, December 31). List of Accredited Biodiesel Producers / Distributors / Projects as of 31 December 2024

22 Department of Energy (Philippines), Renewable Energy Management Bureau, Biomass Energy Management Division. (2024, December 31). List of Accredited Bioethanol Producers / Distributors / Projects as of 31 December 2024.

The country's crude oil import requirements were largely sourced from the Middle East, with Saudi Arabia's hefty 51.4 percent share to total import volume, followed by the United Arab Emirates and Iraq, with shares of 30.9 percent and 13.4 percent, respectively. In contrast, refined petroleum products were largely imported from Asia. South Korea and China emerged as the Philippines' top suppliers, a combined share of 53.4 percent of total refined product import, while Singapore also contributed 14.6 percent share.

Crude oil exports increased by 4.8 percent to 67.0 kTOE compared to last year's 63.9 kTOE. Since operational adjustments necessitated temporary shutdown of Petron's Bataan refinery, it led to the surplus in crude oil that was not processed domestically and instead exported to neighboring countries, such as Thailand. On the other hand, the reduced viability of the Malampaya gas field led to a 9.0 percent drop in condensate exports in 2024, driven by declining indigenous gas reserves and lower production output. In contrast, fulfillment of regional supply contracts boosted the export volume of finished petroleum products by 6.7 percent to 774.8 kTOE despite limited domestic refinery capacity. The Philippines maintained its role as a regional supplier, with key export destinations that included China, Brunei, and Thailand.

Coal imports reached 21.3 million tons of oil equivalent (MTOE) in 2024 or 13.6 percent higher than its volume of 18.6 MTOE in 2023. This growth was primarily driven by the intensified use of coal in power generation. Indonesia continued to serve the Philippines as its dominant coal supplier (99.5 percent share), while the rest were sourced from Australia, China, Vietnam, South Africa, and South Korea. Exports of domestically produced coal climbed by 4.8 percent to 4.2 MTOE, which resulted from sustained high domestic coal output and elevated demand from overseas markets, as the combined demand from China and South Korea, the Philippines' leading export markets, rose by 10.9 percent in 2024.

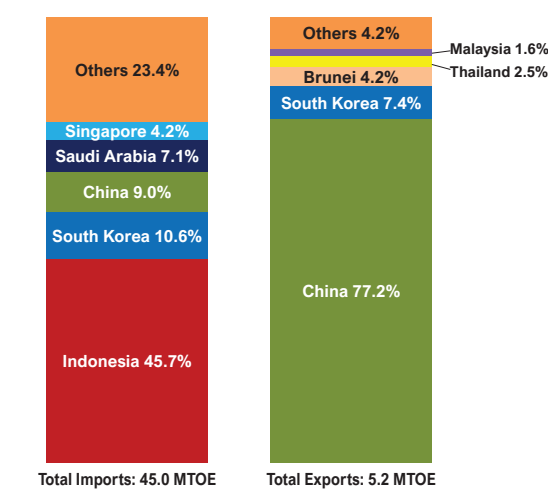


Figure 16: Energy Trade Partners: 2024

LNG imports more than doubled from 643.8 kTOE in 2023 to 1.4 MTOE in 2024. Top import LNG import sources were Australia, which accounted for 39.6 percent share, followed by Nigeria (28.2 percent), and Equatorial Guinea (16.3 percent). The Philippines operated two LNG receiving terminals in 2024: Linseed Field Corporation's floating storage unit (FSU) and onshore regasification and FGen LNG Corporation's floating storage and regasification unit (FSRU). On the other hand, despite higher domestic production



in 2024 vis-à-vis 2023, ethanol import volume rose by 19.6 percent in compliance with mandated blend ratios amidst increased demand from the transport sector.

The country remained as a net importer in 2024, as it recorded a trade in energy goods balance of 40.7 MTOE. This was 9.7 percent higher than its year-ago level and attributed to a sharper rise in energy imports to 45.8 MTOE, which expanded by 9.1 percent or more than double the previous year's 4.5 percent growth in energy exports at 5.2 MTOE (**Figure 16**). Indonesia was the Philippines' leading energy import partner with 45.7 percent share to energy imports due to its dominant role as coal supplier for power generation. This was followed by South Korea (10.6 percent), China (9.0 percent), Saudi Arabia (7.1 percent), and Singapore (4.2 percent). Together, the top five trading partners supplied 76.6 percent of the country's total energy imports in 2024.

On the other hand, the Philippines shipped a total of 5.2 MTOE of combined oil, coal and condensate volume in 2024. China continued to be the primary destination of Philippine energy commodities, particularly coal, as it absorbed 4.0 MTOE or 77.2 percent of total energy exports due to strong trade relations, cost-efficient logistics, and a mutually beneficial trade structure. Other key export markets included South Korea (7.4 percent), Brunei (4.2 percent), Thailand (2.5 percent), and Malaysia (1.6 percent), bringing the top five destinations to a combined 92.8 percent share of total exports.

For international aviation, consumption of jet fuel increased from 1.6 MTOE in 2023 to 1.8 MTOE in 2024, up by 9.6 percent due to continued recovery of global air travel, with more flight routes restored, higher passenger traffic, and a boost from tourism-related activities. Meanwhile, the combined diesel and fuel consumption for international marine bunkers rose to 55.0 kTOE from 52.5 kTOE in 2023, or an increase of 4.7 percent. This growth reflected stronger shipping activity, supported by higher international trade volumes and increased cargo movement through Philippine ports. Both segments posted year-on-year gains, highlighting the sustained rebound of international transport demand after earlier pandemic-induced contractions. The combined volume of coal and other petroleum products was recorded as stock-build up in 2024 stood at 829.1 kTOE.



IV. ENVIRONMENTAL IMPACT

In 2024, total greenhouse gas (GHG) emissions rose by 6.8 percent to 162.1 million tonnes of CO₂ equivalent (MTCO₂e) from 151.8 MTCO₂e in 2023. This increase was largely due to the sustained reliance on fossil fuels, particularly coal and oil, for power generation and transportation. The power sector remained as the largest GHG emitter with 60.1 percent share to the national total, as emissions from coal-fired plants surged by 8.0 percent to 97.5 MTCO₂e in 2024 from the previous year's 90.2 MTCO₂e. **(Table 1).**

GHG emission from the transport sector went up by 5.1 percent to 38.8 MTCO₂e in 2024 from last year's 36.9 MTCO₂e driven by economic growth and increased household consumption. The sector accounted for 23.9 percent of total GHG emissions, which is attributed to its continued dependence on internal combustion engine (ICE) vehicles, which still comprised the largest share of the national vehicle fleet despite aggressive efforts to promote electric mobility.

Table 1: GHG Emission, by Sector: 2023 vs 2024

Sector	CO2 Emission (MTCO ₂ e)		Total NonCO ₂ Emission (MTCO ₂ e)		Total GHG Emission*** (MTCO ₂ e)		Total GHG Emission (% Change)
	2023	2024	2023	2024	2023	2024	2023-2024
Electricity	89.86	97.07	0.36	0.39	90.22	97.46	8.02
Transport	36.67	38.54	0.26	0.27	36.93	38.81	5.10
Industry	12.68	12.89	0.07	0.07	12.75	12.96	1.68
Other Sectors*	10.28	11.08	0.07	0.07	10.35	11.15	7.73
Energy**	1.5	1.75	0.00	0.00	1.51	1.75	16.48
Total	150.99	161.33	0.76	0.80	151.75	162.13	6.84
% Distribution							Change in Distribution
Electricity	56.51	60.17	47.61	48.19	59.45	60.11	0.66
Transport	24.29	23.89	33.92	33.54	24.34	23.94	(0.40)
Industry	8.40	7.99	9.06	8.85	8.40	7.99	(0.41)
Other	6.81	6.86	8.78	8.86	6.82	6.87	0.06
Energy	0.99	1.08	0.62	0.56	0.99	1.08	0.09
Total	100.0	100.0	100.0	100.0	100.0	100.0	

*includes emissions from the services, households, and agriculture

**includes losses incurred in oil refining

*** Updated using GWP Values, Fifth Assessment Report(AR5), and EF based on 2006 IPCC Guidelines (Tier 1)

Industrial GHG emissions reached 13.0 MTCO₂e in 2024 and translated to 8.0 percent of the total GHG emissions. This sluggish uptick of 1.7 percent vis-à-vis 2023 level reflected the modest growth in industrial energy use, as well as increased adoption of cleaner technologies, and gradual shift toward less GHG emissions-intensive industries. Meanwhile, combined emissions from agriculture, services, and households increased by 7.7 percent to 11.1 MTCO₂e from 10.3 MTCO₂e, with a collective contribution of 6.9 percent to the national GHG emission level. Additionally, emissions from refinery operations and own-use energy climbed 16.5 percent, from 1.5 MTCO₂e to 1.8 MTCO₂e.



Table 2: GHG Emission, by Fuel: 2023 vs 2024

Sector	CO2 Emission (MTCO ₂ e)		Total NonCO2 Emission (MTCO ₂ e)		Total GHG Emission*** (MTCO ₂ e)		Total GHG Emission (% Change)
	2023	2024	2023	2024	2023	2024	2023-2024
Oil	55.35	57.53	0.35	0.37	55.70	57.89	3.94
Coal	89.75	96.66	0.40	0.43	90.15	97.10	7.71
Gas	5.9	7.14	0.01	0.01	5.90	7.14	21.05
Total	150.99	161.33	0.76	0.80	151.75	162.13	6.84
% Distribution							Change in Distribution
Oil	36.66	35.66	46.14	45.44	36.71	35.71	(1.00)
Coal	59.44	59.92	53.10	53.69	59.41	59.89	0.48
Gas	3.9	4.42	0.76	3.89	3.89	4.41	0.52
Total	100.0	100.0	100.0	100.0	100.0	100.0	

*** GWP Values and EF based on the Fifth Assessment Report (AR5) and 2006 IPCC Guidelines (Tier 1), respectively

Higher generation output from coal and natural gas-fired power plants in 2024 pushed GHG emissions from coal and natural gas by 7.7 percent and 21.1 percent, respectively, and accounted for an aggregate share of 64.3 percent of total GHG emissions. Oil-related emissions rose 3.9 percent, from 55.7 to 57.9 MTCO₂e, comprising 35.7 percent of the total during the period. (Table 2)

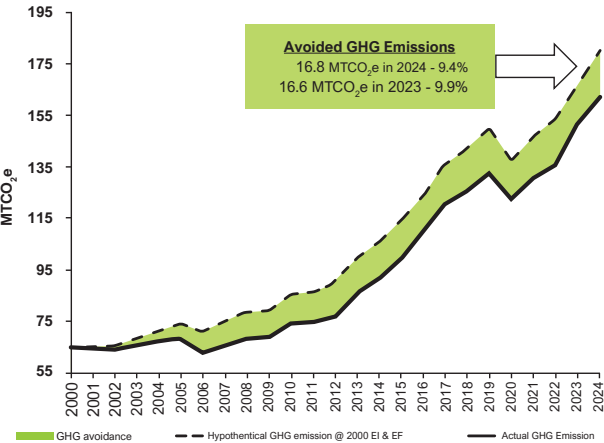


Figure 17. Actual GHG Emission, Hypothetical GHG Emission and GHG Avoidance: 2000 – 2024

Note: Hypothetical GHG Emission is equivalent to Actual GHG Emission plus GHG Emission Avoidance; GHG Base year is CY 2000 GHG Emission Level

The country recorded a 1.1 percent increase in greenhouse gas (GHG) avoidance in 2024, which translated to 16.8 MTCO₂e as illustrated in Figure 19 and Table 3. This was primarily driven by the 10.8 percent hike in avoided emissions through utilization of renewable energy sources, such as geothermal, hydro, wind, solar and biomass, in power generation, from 3.8 MTCO₂e in 2023 to 4.2 MTCO₂e in 2024. In contrast, GHG

avoidance from demand-side management declined by 1.8 percent, from 12.8 MTCO₂e to 12.6 MTCO₂e. Overall, the gains in the supply-side offset the drop in demand-side contributions, resulting in a modest net improvement in national GHG avoidance performance for the year.



Table 3. CO₂ Avoidance from the Mitigation Measures (in ktCO₂e)

GHG Reduction Measures	2023	Reduction Impact* %	2024	Reduction Impact* %	% Change
Demand side	12,805.71	7.61	12,581.48	7.03	-1.75
Efficiency in Electricity Consumption (EEC)	3,806.59	2.26	3,711.29	2.07	-2.50
Efficiency in Fossil Fuel Consumption (EEF)	6,814.23	4.05	6,620.82	3.70	-2.84
Biofuel	2,184.89	1.30	2,249.38	1.26	2.95
CNG/NG	0.00	0.00	0.00	0.00	-2.42
Supply side					
Fuel Diversification in Power Generation @ 2018 GDP & EF**	3,801.20	2.26	4,210.61	2.35	10.77
Total Avoidance (Demand + Supply - EEC)	16,606.91	9.86	16,792.09	9.38	1.12
Actual GHG Emission	151,749.51		162,133.22		6.84
Hypothetical GHG Emission (Actual + Total Avoidance)	168,356.42		178,925.31		6.28

*Refers to the percent reduced emission (Total Avoidance / Hypothetical GHG Emission x 100)

** Includes efficiency in Power Generation and EEC

V. ENERGY – ECONOMY AND ENVIRONMENTAL INDICATORS

The Philippine economy continued to show resilience and steady momentum, posting a slightly faster gross domestic product (GDP) growth of 5.7 percent in 2024 compared to the previous year's 5.5 percent. Despite this modest acceleration, the country ranked as the second fastest growing economy in the region – next to Vietnam and ahead of its other regional peers²³. This underscored the domestic economy's ability to sustain growth amidst persistent external challenges, including subdued global demand, climate-related disruptions, and geopolitical tensions reminiscent of 2023.

The sectoral gross value added (GVA) of industry and services remained the strongest engines of growth for the year, expanding by 6.7 percent and 5.6 percent, respectively. The industry benefitted from recovery in manufacturing and public infrastructure while the services sector remained buoyant with thriving information technology and business process management (IT-BPM) activities²⁴ and domestic tourism²⁵. On the other hand, the agriculture sector contracted by 1.5 percent, weighed down by back-to-back typhoons²⁶ that impacted northern Luzon, coupled with the prevalence of animal diseases that disrupted the sector's value chains during the year.

On the expenditure side, capital formation, which includes public and private construction, was the largest contributor to economic growth as it registered a 7.7 percent expansion in 2024. Meanwhile, government spending grew by 7.3 percent, while

23 2024 Beyond Boundaries Annual Report. (2024). Bangko Sentral ng Pilipinas (BSP). https://www.bsp.gov.ph/Media_And_Research/Annual%20Report/Ann-Rep_2024.pdf

24 IT & Business Process Association Philippines (IBPAP). (2025, January 16). Philippine IT-BPM Industry Caps 2024 with Milestone Achievements. <https://ibpap.org/news-room/21>

25 Department of Tourism. (2025, January 13). Philippines hits record high tourism revenue in 2024. https://beta.tourism.gov.ph/news_and_updates/phl-hits-record-high-tourism-revenue-in-2024/

26 Cagayan's agri damage from 4 typhoons reaches P1.4-B. (n.d.). Philippine News Agency. <https://www.pna.gov.ph/articles/1237983>

household consumption posted a modest increment of 4.9 percent, as persistent food price inflation²⁷ caused by low agricultural output led to tightened consumer spending.

ENERGY-ECONOMY INDICATORS

ENERGY INTENSITY. The country’s overall energy intensity²⁸ slightly went down by 0.5 percent to 3.10 TOE/MPhP reflective of modest gains from stronger energy efficiency standards, technology adoption, and enforcement of regulations. Similarly, oil intensity registered a 1.6 percent decline to 7.47 bbl/PhP in 2024, driven by EV adoption and fleet modernization—and in industry through fuel switching. On the other hand, the amount of electricity use per unit of economic output rose from 5.61 Wh/PhP to 5.71 Wh/PhP due to higher consumption in energy-intensive industries, cold storage, and expanding EV use. These trends underscored the inverse trend between electricity and oil intensities, as electrification led to reduced oil reliance.

By sector, agriculture posted the sharpest rise in energy intensity of 10.9 percent to 0.25 TOE/MPhP in 2024 brought about by increased mechanization, irrigation, and post-harvest processing. Industry and households registered intensity reductions of 2.6 percent and 0.6 percent, respectively, while services recorded a modest 1.6 percent increase, driven by ICT, commercial, and tourism-related demand.

Table 4: Energy Indicators 2023 vs 2024

Indicators	ENERGY	ELECTRICITY	OIL
			
Intensity	TOE/MPhp	Wh/Php	bbl/Php
2023	3.12	5.61	7.60
2024	3.10	5.71	7.47
Per Capita	TOE	kWh	bbl
2023	0.59	1,055.42	1.43
2024	0.61	1,125.56	1.47
Elasticity			
2023	1.30	1.05	0.16
2024	0.91	1.33	0.70

ENERGY ELASTICITY²⁹. In 2024, economic output outperformed energy supply, resulting in an energy elasticity of 0.94 compared to 1.27 in 2023. It signified efficiency improvements, and highlighted the economic contribution of services sector, which was generally less-energy intensive. In contrast, oil elasticity increased to 0.70 and pointed to the rebound in fuel demand from heightened transport activity and industrial use. On the other hand, electricity elasticity climbed further to 1.33 in 2024, which implied that robust electricity demand was fueled by expanding electrification in transport and industry, the proliferation of cold storage facilities, and continued penetration of EVs. These movements highlight a structural transition from oil-based to electricity-based energy use.

27 PBBM admin made every effort to shield Filipinos from high food prices, keeps inflation within target in 2024. (2024, December 17). Department of Finance. <https://www.dof.gov.ph/pbbm-admin-made-every-effort-to-shield-filipinos-from-high-food-prices-keeps-inflation-within-target-in-2024/>

28 Measured in terms of the amount of energy (TOE) used to produce economic output in million pesos of real GDP or MPhP).

29 Elasticity of energy to GDP measures the responsiveness of changes in energy to changes in economic output



ENERGY PER CAPITA. Per capita energy, electricity, and oil use in 2024 rose to 0.61 TOE, 1,125.56 kWh, and 1.47 bbl, respectively, and surpassed their 2023 levels. This upward trend reflected improved access to energy for both daily needs and productive activities, supporting higher living standards and economic participation.

GHG Emission Indicators

Carbon intensity of GDP in 2024 remained relatively at the same level as the previous year's 0.7 tonnes of CO₂ equivalent (tCO₂e) per PHP 100,000. Meanwhile, the carbon intensity of TPES and electricity generation increased by 1.4 percent and 0.4 percent, respectively. These upticks were largely attributed to the growing share of fossil fuels, particularly coal, in the energy and power generation mixes. Consequently, GHG emissions per capita also rose by 5.9 percent, in line with higher overall energy consumption and emissions output. **(Figure 18).**

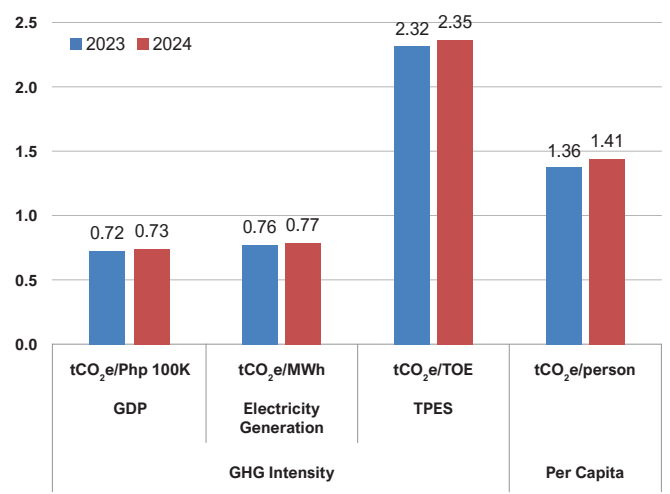


Figure 18. Environmental Emission Indicators: 2023 vs 2024

2024 Energy Balance Table

in thousand tons of oil equivalent (kTOE)

	Coal	Natural Gas	Condensate	Crude	Gasoline	Kerosene	Diesel	Fuel Oil	LPG	Jet A-1	Avgas
Indigenous	8,401.29	1,638.17	180.97	60.19	-	-	-	-	-	-	-
Imports (+)	21,299.76	1,415.18	-	6,097.24	4,372.50	20.57	7,281.93	948.22	1,797.46	1,128.52	3.84
Exports (-)	(4,157.47)	-	(180.97)	(66.95)	(50.86)	-	(32.41)	(22.98)	(2.26)	-	-
International Marine Bunkers (-)	-	-	-	-	-	-	(39.56)	(15.39)	-	-	-
International Civil Aviation (-)	-	-	-	-	-	-	-	-	-	(1,777.21)	-
Stock Change (+/-)	(911.14)	-	-	(79.14)	(26.05)	(2.09)	75.01	(16.72)	(7.08)	(21.82)	0.32
Total Primary Energy Supply	24,632.45	3,053.34	-	6,011.34	4,295.60	18.48	7,284.97	893.13	1,788.12	(670.51)	4.15
Refinery (Crude Run)	-	-	-	(6,409.25)	1,612.92	20.52	2,265.85	55.45	231.56	1,069.04	-
Power Generation	-	-	-	-	-	-	-	-	-	-	-
Fuel Input (-)	(22,403.00)	(2,810.52)	-	-	-	-	(440.29)	(263.37)	-	-	-
Electricity Generation	6,823.67	1,551.74	-	-	-	-	101.93	13.43	-	-	-
Transmission/ Distribution Loss (-)	-	-	-	-	-	-	-	-	-	-	-
Energy Sector Use & Loss (-)	-	(242.85)	-	(388.71)	-	-	-	-	-	-	-
Net Domestic Supply	2,229.45	-	-	(786.63)	5,908.51	39.00	9,110.53	685.22	2,019.68	398.53	4.15
Statistical Difference	-	-	-	-	(17.22)	(3.12)	(139.75)	10.25	(30.62)	37.44	(0.06)
% Statistical Difference	-	-	-	-	-	(7.99)	(1.53)	1.50	(1.52)	9.40	(1.37)
Total Final Energy Consumption	2,229.45	-	-	-	5,925.73	42.12	9,250.28	674.96	2,050.29	361.08	4.21
Industry	2,146.65	-	-	-	-	12.21	996.29	295.46	187.54	-	-
Manufacturing	2,146.65	-	-	-	-	12.02	319.22	244.34	183.37	-	-
Beverages	46.91	-	-	-	-	0.06	46.57	19.29	2.56	-	-
Tobacco	-	-	-	-	-	-	0.21	0.01	2.67	-	-
Coco/Vegetable Oil	26.25	-	-	-	-	0.10	10.63	0.83	6.96	-	-
Sugar	4.85	-	-	-	-	1.89	23.54	0.64	0.04	-	-
Food Processing	262.85	-	-	-	-	5.00	46.21	12.08	80.91	-	-
Textiles/Apparel	4.25	-	-	-	-	-	0.07	2.02	2.05	-	-
Wood Prod/Furniture	-	-	-	-	-	-	4.60	2.28	4.24	-	-
Paper Prod/Printing	77.06	-	-	-	-	0.01	4.88	1.69	1.16	-	-



Naphtha	Asphalt	Other Petroleum Products	Hydro	Geothermal	Solar	Wind	Biomass	Biodiesel	Bioethanol	Electricity	Total
-	-	-	2,715.61	9,277.06	327.66	106.51	7,898.86	218.76	216.63	-	31,041.71
1,030.32	160.46	56.41	-	-	-	-	-	-	228.75	-	45,841.16
(22.98)	-	(643.29)	-	-	-	-	-	-	-	-	(5,180.18)
-	-	-	-	-	-	-	-	-	-	-	(54.95)
-	-	-	-	-	-	-	-	-	-	-	(1,777.21)
94.38	5.87	70.53	-	-	-	-	-	(27.57)	16.42	-	(829.07)
1,101.72	166.33	(516.35)	2,715.61	9,277.06	327.66	106.51	7,898.86	191.20	461.80	-	69,041.47
16.12	-	777.58	-	-	-	-	-	-	-	-	(360.22)
-	-	-	-	-	-	-	-	-	-	-	-
-	-	-	(2,715.61)	(9,277.06)	(327.66)	(106.51)	(562.81)	(8.55)	-	-	(38,915.37)
-	-	-	937.97	927.71	327.66	106.51	124.33	-	-	10,914.95	10,914.95
-	-	-	-	-	-	-	-	-	-	(1,072.63)	(1,072.63)
-	-	-	-	-	-	-	-	-	-	(907.82)	(1,539.36)
1,117.84	166.33	261.23	-	-	-	-	7,336.06	182.64	461.80	8,934.49	38,068.83
(6.85)	30.08	77.43	-	-	-	-	-	-	-	(0.00)	(829.04)
(0.61)	18.08	29.64	-	-	-	-	-	-	-	(0.00)	(2.18)
1,124.69	136.26	183.80	-	-	-	-	7,336.06	182.64	461.80	8,934.50	38,897.87
-	-	-	-	-	-	-	930.83	19.53	-	2,671.87	7,260.39
-	-	-	-	-	-	-	930.83	6.32	-	2,263.29	6,106.04
-	-	-	-	-	-	-	-	0.92	-	40.23	156.54
-	-	-	-	-	-	-	-	0.00	-	2.44	5.33
-	-	-	-	-	-	-	-	0.21	-	86.01	130.99
-	-	-	-	-	-	-	405.32	0.44	-	33.56	470.29
-	-	-	-	-	-	-	514.34	0.95	-	545.63	1,467.96
-	-	-	-	-	-	-	-	0.00	-	70.97	79.36
-	-	-	-	-	-	-	-	0.09	-	10.83	22.05
-	-	-	-	-	-	-	-	0.10	-	70.98	155.88



2024 Energy Balance Table

in thousand tons of oil equivalent (kTOE)

	Coal	Natural Gas	Condensate	Crude	Gasoline	Kerosene	Diesel	Fuel Oil	LPG	Jet A-1	Avgas
Chemicals Except Fertilizer	33.60	-	-	-	-	4.30	37.48	42.74	22.96	-	-
Fertilizer	24.01	-	-	-	-	-	2.58	-	0.52	-	-
Lube Refining	-	-	-	-	-	-	-	-	-	-	-
Glass/Glass Products	-	-	-	-	-	-	7.46	51.58	1.58	-	-
Cement	1,211.05	-	-	-	-	-	24.30	24.67	0.67	-	-
Other Non-Metic Minerals	4.56	-	-	-	-	-	1.49	0.62	0.05	-	-
Basic Metal	225.18	-	-	-	-	0.00	13.87	74.88	14.99	-	-
Machinery/Equipment	-	-	-	-	-	0.67	49.57	6.07	19.40	-	-
Other Manufacturing	203.47	-	-	-	-	-	44.64	4.79	21.58	-	-
Rubber/ Rubber Products	22.61	-	-	-	-	-	1.13	0.13	1.03	-	-
Mining	-	-	-	-	-	0.00	419.77	49.37	-	-	-
Construction	-	-	-	-	-	0.19	257.31	1.75	4.17	-	-
Transport	-	-	-	-	5,921.36	-	6,420.35	231.45	-	361.08	4.21
Railway	-	-	-	-	-	-	1.08	-	-	-	-
Road Transport	-	-	-	-	5,720.14	-	5,979.85	-	-	-	-
Water Transport	-	-	-	-	201.22	-	439.42	231.45	-	-	-
Domestic Air Transport	-	-	-	-	-	-	-	-	-	361.08	4.21
Households	-	-	-	-	-	29.90	-	-	1,343.36	-	-
Services	-	-	-	-	-	-	1,690.91	147.16	519.40	-	-
Agriculture	-	-	-	-	4.37	0.01	142.73	0.89	-	-	-
Agri-Industry	-	-	-	-	3.05	0.00	27.62	-	-	-	-
Agri Crops Product	-	-	-	-	3.04	0.00	25.95	-	-	-	-
Livestock/Poultry	-	-	-	-	0.01	-	1.67	-	-	-	-
Agri Services	-	-	-	-	-	-	-	-	-	-	-
Forestry	-	-	-	-	-	-	-	-	-	-	-
Fishery	-	-	-	-	1.33	0.00	115.10	0.89	-	-	-
Non-Energy Use	82.80	-	-	-	-	-	-	-	-	-	-
Self-sufficiency											



Naphtha	Asphalt	Other Petroleum Products	Hydro	Geothermal	Solar	Wind	Biomass	Biodiesel	Bioethanol	Electricity	Total
-	-	-	-	-	-	-	-	0.75	-	92.62	234.44
-	-	-	-	-	-	-	-	0.05	-	17.98	45.13
-	-	-	-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	0.14	-	8.20	68.96
-	-	-	-	-	-	-	-	0.49	-	33.87	1,295.06
-	-	-	-	-	-	-	-	0.03	-	6.62	13.37
-	-	-	-	-	-	-	-	0.28	-	232.07	561.27
-	-	-	-	-	-	-	-	0.96	-	952.93	1,029.60
-	-	-	-	-	-	-	11.17	0.90	-	57.58	344.13
-	-	-	-	-	-	-	-	0.02	-	0.74	25.66
-	-	-	-	-	-	-	-	8.17	-	119.00	596.31
-	-	-	-	-	-	-	-	5.04	-	289.59	558.05
-	-	-	-	-	-	-	-	126.78	461.80	21.19	13,548.23
-	-	-	-	-	-	-	-	0.02	-	10.09	11.19
-	-	-	-	-	-	-	-	119.42	461.80	11.10	12,292.32
-	-	-	-	-	-	-	-	7.34	-	-	879.42
-	-	-	-	-	-	-	-	-	-	-	365.29
-	-	-	-	-	-	-	6,068.80	-	-	3,543.02	10,985.08
-	-	-	-	-	-	-	336.42	33.58	-	2,410.40	5,137.87
-	-	-	-	-	-	-	-	2.75	-	288.01	438.75
-	-	-	-	-	-	-	-	0.54	-	213.09	244.31
-	-	-	-	-	-	-	-	0.51	-	133.93	163.43
-	-	-	-	-	-	-	-	0.03	-	67.87	69.58
-	-	-	-	-	-	-	-	-	-	11.30	11.30
-	-	-	-	-	-	-	-	-	-	1.10	1.10
-	-	-	-	-	-	-	-	2.20	-	73.82	193.35
1,124.69	136.26	183.80	-	-	-	-	-	-	-	-	1,527.54
											44.96



2023 Energy Balance Table

in thousand tons of oil equivalent (kTOE)

	Coal	Natural Gas	Condensate	Crude	Gasoline	Kerosene	Diesel	Fuel Oil	LPG	Jet A-1	Avgas
Indigenous	8,051.68	1,878.57	198.82	67.36	-	-	-	-	-	-	-
Imports (+)	18,755.26	643.85	-	6,381.49	4,099.16	21.19	6,879.23	1,016.38	1,726.21	1,062.93	4.32
Exports (-)	(3,967.10)	-	(198.82)	(63.86)	(11.56)	-	(95.92)	(39.53)	-	-	-
International Marine Bunkers (-)	-	-	-	-	-	-	(31.15)	(21.33)	-	-	-
International Civil Aviation (-)	-	-	-	-	-	-	-	-	-	(1,620.84)	-
Stock Change (+/-)	114.14	-	-	31.70	(76.19)	1.28	(183.97)	(41.12)	5.72	(35.74)	(0.42)
Total Primary Energy Supply	22,953.99	2,522.42	-	6,416.69	4,011.41	22.47	6,568.18	914.41	1,731.93	(593.65)	3.90
Refinery (Crude Run)	-	-	-	(6,847.30)	1,628.31	22.13	2,590.38	45.95	237.20	947.35	-
Power Generation	-	-	-	-	-	-	-	-	-	-	-
Fuel Input (-)	(20,886.86)	(2,466.62)	-	-	-	-	(385.89)	(311.22)	-	-	-
Electricity Generation	6,341.69	1,433.20	-	-	-	-	96.43	15.66	-	-	-
Transmission/ Distribution Loss (-)	-	-	-	-	-	-	-	-	-	-	-
Energy Sector Use & Loss (-)	-	(55.79)	-	(450.83)	-	-	-	-	-	-	-
Net Domestic Supply	2,067.13	-	-	(881.4)	5,639.72	44.6	8,772.7	649.1	1,969.1	353.7	3.9
Statistical Difference	-	-	-	-	39.44	(2.54)	(36.86)	(18.17)	(34.03)	(31.04)	0.01
% Statistical Difference	-	-	-	-	-	(5.69)	(0.42)	(2.80)	(1.73)	(8.78)	0.30
Total Final Energy Consumption	2,067.13	-	-	-	5,600.28	47.13	8,809.53	667.32	2,003.16	384.74	3.89
Industry	1,987.07	-	-	-	-	15.26	1,085.14	296.45	198.63	-	-
Manufacturing	1,987.07	-	-	-	-	15.11	337.33	245.77	193.09	-	-
Food, beverages and tobacco	349.46	-	-	-	-	7.57	127.37	36.62	93.75	-	-
Beverages	50.40	-	-	-	-	0.04	45.22	21.92	3.20	-	-
Tobacco	-	-	-	-	-	-	0.17	0.01	3.08	-	-
Coco/Vegetable Oil	24.46	-	-	-	-	0.08	11.37	0.02	0.18	-	-
Sugar	10.95	-	-	-	-	1.59	19.33	0.37	0.04	-	-
Food Processing	263.65	-	-	-	-	5.85	51.27	14.29	87.24	-	-
Textile and leather	16.04	-	-	-	-	-	0.23	1.41	2.79	-	-
Textiles/Apparel	16.04	-	-	-	-	-	0.23	1.41	2.79	-	-
Wood and wood products	-	-	-	-	-	-	1.48	-	6.39	-	-
Wood ProdFurniture	-	-	-	-	-	-	1.48	-	6.39	-	-
Pulp, paper and print	202.22	-	-	-	-	0.01	5.61	1.49	1.26	-	-
Paper Prod/Printing	202.22	-	-	-	-	0.01	5.61	1.49	1.26	-	-



Naphtha	Asphalt	Other Petroleum Products	Hydro	Geothermal	Solar	Wind	Biomass	Biodiesel	Bioethanol	Electricity	Total
-	-	-	2,560.91	9,226.30	218.75	112.48	7,818.93	186.04	213.99	-	30,533.82
800.15	100.40	345.02	-	-	-	-	-	-	191.29	-	42,026.89
(129.43)	-	(449.59)	-	-	-	-	-	-	-	-	(4,955.81)
-	-	-	-	-	-	-	-	-	-	-	(52.48)
-	-	-	-	-	-	-	-	-	-	-	(1,620.84)
(44.74)	4.35	(89.05)	-	-	-	-	-	(4.60)	31.85	-	(286.79)
625.98	104.76	(193.62)	2,560.91	9,226.30	218.75	112.48	7,818.93	181.43	437.14	-	65,644.80
133.04	-	867.85	-	-	-	-	-	-	-	-	(375.10)
-	-	-	-	-	-	-	-	-	-	-	-
-	-	-	(2,560.91)	(9,226.30)	(218.75)	(112.48)	(548.89)	(7.61)	-	-	(36,725.52)
-	-	-	884.54	922.63	218.75	112.48	121.14	-	-	10,146.51	10,146.51
-	-	-	-	-	-	-	-	-	-	(1,014.00)	(1,014.00)
-	-	-	-	-	-	-	-	-	-	(894.51)	(1,401.14)
759.0	104.8	674.2	-	-	-	-	7,270.0	173.8	437.1	8,237.99	36,275.54
(93.25)	(7.80)	322.44	-	-	-	-	-	-	-	(0.00)	(743.25)
(12.29)	(7.44)	47.82	-	-	-	-	-	-	-	(0.00)	(2.05)
852.26	112.55	351.79	-	-	-	-	7,270.04	173.82	437.14	8,237.99	37,018.79
-	-	-	-	-	-	-	928.12	21.41	-	2,535.91	7,067.99
-	-	-	-	-	-	-	928.12	6.66	-	2,148.11	5,861.26
-	-	-	-	-	-	-	917.00	2.51	-	671.85	2,206.13
-	-	-	-	-	-	-	-	0.89	-	38.19	159.87
-	-	-	-	-	-	-	-	0.00	-	2.31	5.59
-	-	-	-	-	-	-	-	0.22	-	81.64	117.97
-	-	-	-	-	-	-	403.50	0.38	-	31.85	468.02
-	-	-	-	-	-	-	513.49	1.01	-	517.87	1,454.68
-	-	-	-	-	-	-	-	0.00	-	67.35	87.82
-	-	-	-	-	-	-	-	0.00	-	67.35	87.82
-	-	-	-	-	-	-	-	0.03	-	10.28	18.18
-	-	-	-	-	-	-	-	0.03	-	10.28	18.18
-	-	-	-	-	-	-	-	0.11	-	67.37	278.07
-	-	-	-	-	-	-	-	0.11	-	67.37	278.07

2023 Energy Balance Table

in thousand tons of oil equivalent (kTOE)

	Coal	Natural Gas	Condensate	Crude	Gasoline	Kerosene	Diesel	Fuel Oil	LPG	Jet A-1	Avgas
Chemical and petrochemical	60.97	-	-	-	-	5.20	42.15	21.74	28.00	-	-
Chemicals Except Fertilizer	32.32	-	-	-	-	5.20	39.80	21.74	27.93	-	-
Fertilizer	28.65	-	-	-	-	-	2.34	-	0.07	-	-
Lube Refining	-	-	-	-	-	-	-	-	-	-	-
Non-metallic minerals	916.84	-	-	-	-	-	37.75	72.49	2.03	-	-
Glass/Glass Products	-	-	-	-	-	-	7.24	48.63	1.37	-	-
Cement	905.79	-	-	-	-	-	28.93	23.15	0.61	-	-
Other Non-Metic Minerals	11.06	-	-	-	-	-	1.59	0.70	0.05	-	-
Iron and steel	224.98	-	-	-	-	0.99	15.01	91.33	15.15	-	-
Basic Metal	224.98	-	-	-	-	0.99	15.01	91.33	15.15	-	-
Machinery	-	-	-	-	-	1.33	71.17	17.02	19.71	-	-
Machinery/Equipment	-	-	-	-	-	1.33	71.17	17.02	19.71	-	-
Not elsewhere specified	216.57	-	-	-	-	-	36.56	3.67	24.03	-	-
Other Manufacturing	200.63	-	-	-	-	-	34.90	3.52	23.22	-	-
Rubber/ Rubber Products	15.94	-	-	-	-	-	1.66	0.15	0.81	-	-
Mining	-	-	-	-	-	0.00	396.83	49.20	-	-	-
Construction	-	-	-	-	-	0.16	350.98	1.48	5.54	-	-
TRANSPORT	-	-	-	-	5,589.82	-	6,111.32	219.35	-	384.74	3.89
Railway	-	-	-	-	-	-	2.13	-	-	-	-
Road Transport	-	-	-	-	5,415.67	-	5,623.20	-	-	-	-
Water Transport	-	-	-	-	174.15	-	485.99	219.35	-	-	-
Domestic Air Transport	-	-	-	-	-	-	-	-	-	384.74	3.89
HOUSEHOLDS	-	-	-	-	-	31.87	-	-	1,317.67	-	-
SERVICES	-	-	-	-	-	-	1,479.65	150.81	486.86	-	-
AGRICULTURE	-	-	-	-	10.46	-	133.42	0.70	-	-	-
Agri-Industry	-	-	-	-	9.14	-	31.80	-	-	-	-
Agri Crops Product	-	-	-	-	9.13	-	29.92	-	-	-	-
Livestock/Poultry	-	-	-	-	0.01	-	1.88	-	-	-	-
Agri Services	-	-	-	-	-	-	-	-	-	-	-
Forestry	-	-	-	-	-	-	-	-	-	-	-
Fishery	-	-	-	-	1.33	-	101.62	0.70	-	-	-
NON-ENERGY USE	80.06	-	-	-	-	-	-	-	-	-	-
Self-sufficiency											



Naphtha	Asphalt	Other Petroleum Products	Hydro	Geothermal	Solar	Wind	Biomass	Biodiesel	Bioethanol	Electricity	Total
-	-	-	-	-	-	-	-	0.83	-	104.98	263.86
-	-	-	-	-	-	-	-	0.79	-	87.91	215.69
-	-	-	-	-	-	-	-	0.05	-	17.06	48.17
-	-	-	-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	0.83	-	104.98	263.86
-	-	-	-	-	-	-	-	0.14	-	7.79	65.17
-	-	-	-	-	-	-	-	0.57	-	32.15	991.20
-	-	-	-	-	-	-	-	0.03	-	6.29	19.72
-	-	-	-	-	-	-	-	0.83	-	104.98	263.86
-	-	-	-	-	-	-	-	0.30	-	220.26	568.02
-	-	-	-	-	-	-	-	1.40	-	904.44	1,015.08
-	-	-	-	-	-	-	-	1.40	-	904.44	1,015.08
-	-	-	-	-	-	-	11.12	0.72	-	55.35	348.01
-	-	-	-	-	-	-	11.12	0.69	-	54.65	328.73
-	-	-	-	-	-	-	-	0.03	-	0.70	19.28
-	-	-	-	-	-	-	-	7.83	-	112.94	566.80
-	-	-	-	-	-	-	-	6.93	-	274.85	639.93
-	-	-	-	-	-	-	-	120.58	437.14	13.21	12,880.05
-	-	-	-	-	-	-	-	0.04	-	9.76	11.93
-	-	-	-	-	-	-	-	114.25	437.14	3.45	11,593.71
-	-	-	-	-	-	-	-	6.29	-	-	885.78
-	-	-	-	-	-	-	-	-	-	-	388.63
-	-	-	-	-	-	-	6,007.79	-	-	3,178.67	10,536.01
-	-	-	-	-	-	-	334.12	29.19	-	2,255.86	4,736.50
-	-	-	-	-	-	-	-	2.63	-	254.35	401.57
-	-	-	-	-	-	-	-	0.63	-	188.19	229.75
-	-	-	-	-	-	-	-	0.59	-	118.27	157.91
-	-	-	-	-	-	-	-	0.04	-	59.94	61.86
-	-	-	-	-	-	-	-	-	-	9.98	9.98
-	-	-	-	-	-	-	-	-	-	0.97	0.97
-	-	-	-	-	-	-	-	2.01	-	65.19	170.84
852.26	112.55	351.79	-	-	-	-	-	-	-	-	1,396.67
											46.51





DEPARTMENT OF ENERGY



BAGONG PILIPINAS



2024

KEY ENERGY

STATISTICS



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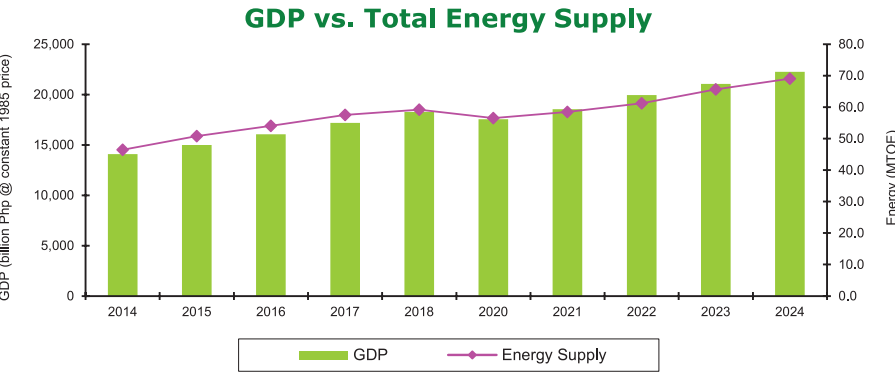


Energy and Economic Indicators

	2014	2015	2016	2017	2018	2019
GDP (in billion pesos: at constant 2018 prices)	14,096.0	14,990.9	16,062.7	17,176.0	18,265.2	19,382.8
Total Final Energy Consumption (in MTOE)	28.5	31.0	33.5	35.5	35.7	36.3
Total Primary Energy Supply (in MTOE)	46.4	50.8	54.1	57.5	59.2	59.3
Population (in million)	99.9	100.8	102.5	104.2	105.8	107.3
Forex (in Pesos/USD)	44.6	47.2	49.8	49.9	52.7	50.7
Average Crude Price (in USD / barrel)	97.0	50.9	42.2	54.2	69.4	66.8

	2020	2021	2022	2023	2024	AAGR*
GDP (in billion pesos: at constant 2018 prices)	17,537.8	18,540.1	19,945.6	21,046.4	22,244.4	4.7%
Total Final Energy Consumption (in MTOE)	32.6	35.0	35.9	37.0	38.9	3.2%
Total Primary Energy Supply (in MTOE)	56.5	58.5	61.2	65.6	69.0	4.0%
Population (in million)	109.2	110.1	110.9	111.8	112.8	1.2%
Forex (in Pesos/USD)	48.0	50.8	56.1	55.6	58.0	2.7%
Average Crude Price (in USD / barrel)	49.8	69.2	70.2	82.1	79.6	-2.0%

*AAGR - Average Annual Growth Rate



Sources:
Gross Domestic Product (GDP), Population -National Accounts, Philippine Statistical Authority (Rebased 2018)
Foreign Exchange Rate - Bangko Sentral ng Pilipinas (BSP)
Energy Supply - Policy Formulation and Research Division (PFRD), DOE
Crude Oil Price - Oil Industry Management Bureau (OIMB), DOE



Indicator	2014	2015	2016	2017	2018	2019
Intensity						
Energy to GDP* (TOE/Php 1M)	3.29	3.39	3.37	3.35	3.24	3.06
Oil to GDP (BBL/Php 100,000)	7.77	8.78	8.88	8.90	8.50	7.70
Electricity to GDP (Wh/Php)	5.48	5.50	5.65	5.49	5.46	5.47
Elasticity						
Energy to GDP	0.68	1.48	0.90	0.91	0.47	0.03
Oil to GDP	0.62	3.17	1.17	1.02	0.25	0.62
Electricity to GDP	0.42	1.05	1.42	0.57	0.90	1.03
Energy Per Capita (TOE/person)	0.46	0.50	0.53	0.55	0.56	0.55

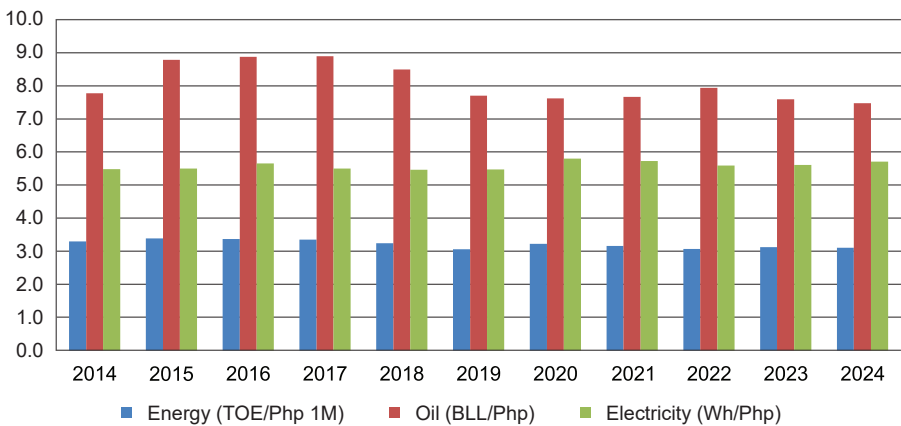
Indicator	2020	2021	2022	2023	2024	AAGR*
Intensity						
Energy to GDP* (TOE/Php 1M)	3.22	3.15	3.07	3.12	3.10	-0.6%
Oil to GDP (BBL/Php 100,000)	7.62	7.67	7.94	7.60	7.47	-0.4%
Electricity to GDP (Wh/Php)	5.80	5.72	5.59	5.61	5.71	0.4%
Elasticity						
Energy to GDP	0.50	0.61	0.63	1.30	0.91	2.9%
Oil to GDP	1.10	1.12	1.51	0.16	0.70	1.3%
Electricity to GDP	0.42	0.75	0.67	1.05	1.33	12.3%
Energy Per Capita (TOE/person)	0.52	0.53	0.55	0.59	0.61	2.8%

*AAGR - Average Annual Growth Rate

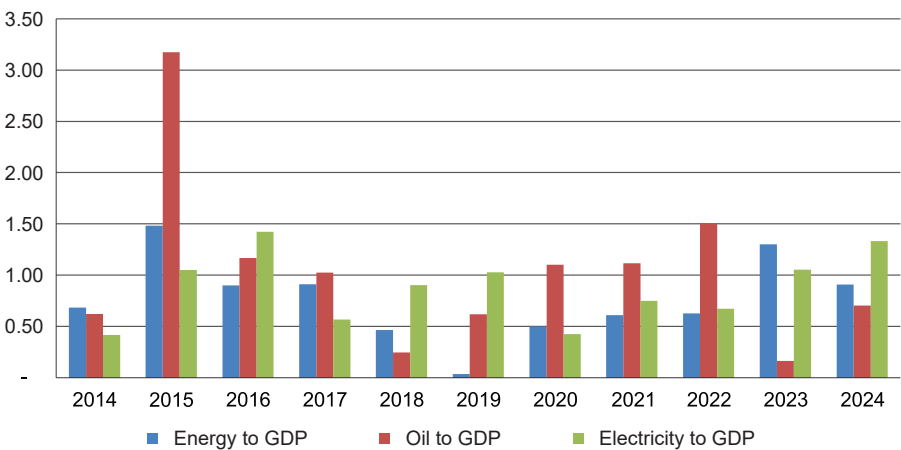
** GDP Rebased 2018 @ constant price



Energy Intensity



Elasticity



Energy and Environment



GHG Emission by Sector and Activity

MTCO₂e⁽¹⁾

Sector and Activity	2014	2015	2016	2017	2018	2019
Industry	12.88	13.20	15.29	16.63	14.22	13.18
Transport	26.45	30.60	33.11	34.19	35.38	36.63
Others ⁽²⁾	7.15	7.06	8.59	10.16	10.63	11.27
Electricity Generation	43.63	47.49	51.61	59.00	64.60	70.32
Energy ⁽³⁾	1.11	0.96	0.66	0.71	0.75	1.05
TOTAL	91.22	99.30	109.25	120.69	125.58	132.45

Sector and Activity	2020	2021	2022	2023	2024	AAGR*
Industry	11.27	12.50	12.94	12.75	12.96	0.1%
Transport	28.26	31.53	35.42	36.93	38.81	3.9%
Others ⁽²⁾	11.33	12.13	9.88	10.35	11.15	4.5%
Electricity Generation	70.92	73.88	76.28	90.22	97.46	8.4%
Energy ⁽³⁾	0.41	0.40	1.16	1.51	1.75	4.7%
TOTAL	122.20	130.45	135.68	151.75	162.13	5.9%

Notes:

(1) Million tons of CO₂ Equivalent (MTCO₂e)

(2) includes Household, Services and Agriculture Sectors

(3) includes Oil refining, Electricity and other Energy sector own use and losses

*average annual growth rate

GHG Emission by Fuel Type

MTCO₂e

Fuel type	2014	2015	2016	2017	2018	2019
Liquid Fossils (Oil)	42.56	47.15	50.47	52.47	53.06	55.31
Solid Fossils (Coal)	41.56	45.48	51.13	60.67	64.10	68.66
Gaseous Fossil (Natural Gas)	7.10	6.68	7.65	7.55	8.42	8.48
TOTAL	91.22	99.30	109.25	120.69	125.58	132.45

Fuel type	2020	2021	2022	2023	2024	AAGR*
Liquid Fossils (Oil)	45.88	49.81	54.40	55.70	57.89	3.1%
Solid Fossils (Coal)	68.63	74.05	75.17	90.15	97.10	8.9%
Gaseous Fossil (Natural Gas)	7.69	6.60	6.11	5.90	7.14	0.1%
TOTAL	122.20	130.45	135.68	151.75	162.13	5.9%

*average annual growth rate



Environmental Emission Indicators

GHG emission is expressed in carbon dioxide equivalent (CO₂e) which accounts for the global warming potential (GWP) of methane (CH₄) and nitrous oxide (N₂O), as prescribed by the Inter-governmental Panel on Climate Change (IPCC). GWP is the ratio of the warming resulting from the emission of one kilogram of a greenhouse gas to that of one kilogram emission of CO₂ over a fixed period of time (i.e. CH₄ and N₂O GWP is 21 times and 310 times the CO₂ emission, respectively).

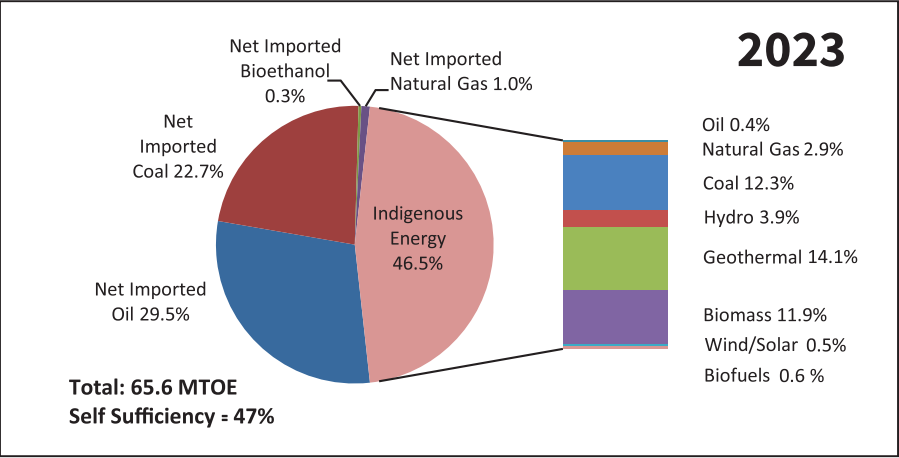
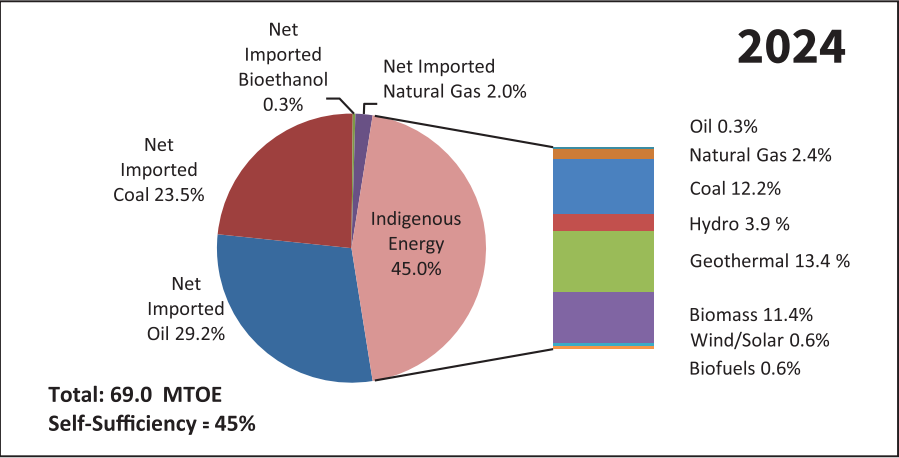
Indicators	2014	2015	2016	2017	2018	2019
GHG emission-to-GDP ratio						
(tCO ₂ e/PhP 100K, 2000=100)	0.65	0.66	0.68	0.70	0.69	0.68
GHG emission per capita						
(tCO ₂ e/person)	0.91	0.98	1.07	1.16	1.19	1.23
GHG emission per Electricity Generation						
(tCO ₂ e/MWh)	0.56	0.58	0.57	0.63	0.65	0.66
GHG emission per Oil consumption						
(tCO ₂ e/TOE)	2.75	2.55	2.58	2.57	2.61	2.85
GHG emission per TPES						
(tCO ₂ e/TOE)	1.96	1.95	2.02	2.10	2.12	2.23

Indicators	2020	2021	2022	2023	2024	AAGR*
GHG emission-to-GDP ratio						
(tCO ₂ e/PhP 100K, 2000=100)	0.70	0.70	0.68	0.72	0.73	1.2%
GHG emission per capita						
(tCO ₂ e/person)	1.12	1.19	1.22	1.36	1.44	4.6%
GHG emission per Electricity Generation						
(tCO ₂ e/MWh)	0.70	0.70	0.68	0.76	0.77	3.1%
GHG emission per Oil consumption						
(tCO ₂ e/TOE)	2.71	2.80	2.66	2.73	2.73	-0.1%
GHG emission per TPES						
(tCO ₂ e/TOE)	2.16	2.23	2.22	2.32	2.35	1.8%

**average annual growth rate*



Total Primary Energy Supply Mix



Total Primary Energy Supply (TPES) and Self-Sufficiency Level

In kTOE

	2014	2015	2016	2017	2018	2019
Indigenous Energy	26,606	26,881	29,405	29,515	29,977	30,906
Oil	849	715	702	622	594	523
Natural Gas	3,036	2,854	3,270	3,226	3,601	3,626
Coal	4,012	3,894	5,917	6,298	6,204	7,258
Hydro	2,275	2,157	2,019	2,393	2,336	1,998
Geothermal	8,863	9,496	9,519	8,831	8,973	9,192
Biomass	7,356	7,431	7,494	7,651	7,725	7,736
Wind	13	64	84	94	99	90
Solar ^(a)	1	12	94	103	107	107
Biodiesel	134	164	178	167	168	178
Bioethanol	66	94	127	131	170	199
Net Imported Energy	19,833	23,929	24,677	27,985	29,218	28,414
Oil	13,021	16,032	17,337	18,589	18,879	18,000
Coal	6,630	7,721	7,169	9,177	10,145	10,224
Bioethanol	182	176	172	219	194	190
TPES	46,439	50,810	54,082	57,499	59,195	59,320
Renewable Energy (RE)	18,891	19,594	19,687	19,588	19,772	19,690
Green Energy (RE + Natural Gas)	21,927	22,448	22,957	22,814	23,373	23,316
Self Sufficiency (%)	57	53	54	51	51	52

	2020	2021	2022	2023	2024	AAGR*
Indigenous Energy	29,676	29,838	30,422	30,534	31,042	1.6%
Oil	456	392	358	266	241	-11.8%
Natural Gas	3,288	2,820	2,612	1,879	1,638	-6.0%
Coal	6,836	7,414	7,631	8,052	8,401	7.7%
Hydro	1,790	2,287	2,510	2,561	2,716	1.8%
Geothermal	9,249	8,613	8,963	9,226	9,277	0.5%
Biomass	7,563	7,721	7,731	7,819	7,899	0.7%
Wind	88	109	89	112	107	23.3%
Solar ^(a)	118	126	157	219	328	44.5%
Biodiesel	131	157	166	186	219	5.0%
Bioethanol	155	200	206	214	217	12.5%
Net Imported Energy	26,827	28,635	30,824	35,111	38,000	6.7%
Oil	15,922	16,954	19,163	19,346	20,136	4.5%
Coal	10,710	11,499	11,445	14,902	16,231	9.4%
Bioethanol	194	183	216	219	218	1.8%
Natural Gas ^(b)	-	-	-	644	1,415	119.8%
TPES	56,502	58,473	61,246	65,645	69,041	4.0%
Renewable Energy (RE)	19,290	19,395	20,038	20,556	20,979	1.1%
Green Energy (RE + Natural Gas)	22,578	22,215	22,650	23,078	24,032	0.9%
Self Sufficiency (%)	53	51	50	46.5	45.0	

*average annual growth rate

a) AAGR from 2015 to 2024

b) AAGR from 2023 to 2024



Energy Consumption



Total Final Energy Consumption, by Sector and Fuel Type*

In kTOE

	2014	2015	2016	2017	2018	2019
Industry	6,529	6,750	7,449	7,925	7,523	7,306
Coal	2,261	2,218	2,677	3,008	2,411	2,217
Natural Gas ^(a)	77	50	65	53	59	62
Oil	1,206	1,382	1,458	1,470	1,469	1,381
Diesel	535	587	656	750	693	754
Fuel Oil	527	637	653	613	578	453
LPG	128	141	130	85	196	173
Kerosene	16	18	19	22	2	2
Biomass	1,131	1,152	1,164	1,181	1,199	1,207
Ricehull	48	49	49	50	51	51
Fuelwood	170	172	173	176	179	180
Bagasse	489	499	504	511	519	522
Agriwaste	404	412	416	423	429	432
Animal Waste	20	21	21	21	21	22
Biodiesel	11	12	13	14	13	15
Electricity	1,843	1,936	2,074	2,199	2,372	2,424
Transport	9,133	10,557	11,425	11,824	12,239	12,698
Natural Gas***	0.1	-	-	-	-	-
Oil	8,781.9	10,151.0	10,986.1	11,351.8	11,752.7	12,181.1
Gasoline	3,181	3,652	4,020	4,370	4,504	4,870
Diesel	4,839	5,841	6,156	6,195	6,452	6,479
Fuel Oil	178	183	203	265	203	196
Auto LPG ^(b)	42	30	13	10	10	9
Aviation fuel	542	446	594	512	584	626
Biodiesel	96	116	121	123	127	127
Bioethanol	246	281	309	339	350	380
Electricity	10	9	9	10	10	10



	2014	2015	2016	2017	2018	2019
Households	8,488	8,731	9,035	9,192	9,431	9,711
Oil	862	973	1,122	1,159	1,255	1,312
LPG	769	888	1,043	1,083	1,176	1,240
Kerosene	93	85	79	75	79	72
Biomass	5,823	5,802	5,709	5,731	5,746	5,772
Fuelwood	4,290	4,259	4,131	4,119	4,107	4,100
Charcoal	946	973	1,027	1,067	1,110	1,157
Agriwaste	588	570	551	545	529	515
Electricity	1,803	1,956	2,204	2,303	2,430	2,627
Services	3,397	3,370	3,865	4,404	4,668	4,936
Oil	1,432	1,292	1,632	2,074	2,223	2,360
Diesel	1,053	843	1,058	1,397	1,568	1,706
Fuel Oil	113	138	201	146	148	158
LPG	266	310	374	532	507	495
Biomass	332	337	340	345	350	353
Ricehull	7	6	6	7	7	7
Charcoal	162	164	166	168	171	172
Fuelwood	164	166	168	170	173	174
Biodiesel	20	14	21	28	30	33
Electricity	1,613	1,727	1,872	1,958	2,065	2,191
Agriculture	354	401	450	515	439	473
Oil	172	194	229	290	208	229
Gasoline	5	8	7	12	7	9
Kerosene ^(c)	0	0	1	0	0	0
Diesel	165	182	212	270	200	218
Fuel Oil	2	3	8	7	1	2
Biodiesel	3	4	4	5	4	4
Electricity	178	203	218	220	227	239
Non-Energy Use	605	1,179	1,306	1,613	1,423	1,137
Oil	450	1,047	1,129	1,458	1,261	996
Coal	154	132	177	155	162	141
Total	28,506	30,988	33,530	35,474	35,723	36,260



	2020	2021	2022	2023	2024	AAGR*
Industry	6,320	6,821	7,107	7,068	7,260	1.1%
Coal	1,629	1,950	1,855	1,987	2,147	-0.5%
Natural Gas ^(a)	37	0.3	-	-	-	-53.7%
Oil	1,537	1,558	1,824	1,595	1,492	2.1%
Diesel	830	888	1,200	1,085	996	6.4%
Fuel Oil	512	450	411	296	295	-5.6%
LPG	178	200	202	199	188	3.9%
Kerosene	17	20	11	15	12	-2.7%
Biomass	905	923	924	928	931	-1.9%
Ricehull	38	39	39	40	40	-1.9%
Fuelwood	135	137	135	135	134	-2.4%
Bagasse	392	400	401	404	405	-1.9%
Agriwaste	324	331	331	334	335	-1.9%
Animal Waste	16	17	17	17	17	-1.9%
Biodiesel	13	15	24	21	20	5.7%
Electricity	2,198	2,375	2,480	2,536	2,672	3.8%
Transport	9,839	10,982	12,322	12,880	13,548	0.2%
Natural Gas***	-	-	-	-	-	-
Oil	9,411	10,503	11,790	12,309	12,938	4.0%
Gasoline	4,145.2	4,714.0	5,188.5	5,589.8	5,921.4	6.4%
Diesel	4,913.5	5,401.1	5,935.8	6,111.3	6,420.4	2.9%
Fuel Oil	128.8	190.9	322.3	219.4	231.5	2.7%
Auto LPG ^(b)	4.4	2.8	0.2	-	-	-47.9%
Aviation fuel	218.7	194.1	342.9	388.6	365.3	-3.9%
Biodiesel	97	101	117	121	127	2.9%
Bioethanol	324	369	406	437	462	6.5%
Electricity	7	9	9	13	21	8.3%



	2020	2021	2022	2023	2024	AAGR*
Households	10,028	10,179	10,310	10,536	10,985	2.6%
Oil	1,238	1,266	1,311	1,350	1,373	4.8%
LPG	1,187	1,215	1,265	1,318	1,343	5.7%
Kerosene	51	51	46	32	30	-10.7%
Biomass	5,842	5,904	5,962	6,008	6,069	0.4%
Fuelwood	4,121	4,136	4,145	4,145	4,153	-0.3%
Charcoal	1,209	1,263	1,317	1,372	1,431	4.2%
Agriwaste	512	506	500	491	484	-1.9%
Electricity	2,949	3,008	3,037	3,179	3,543	7.0%
Services	4,606	4,848	4,452	4,736	5,138	4.2%
Oil	2,462	2,662	2,005	2,117	2,357	5.1%
Diesel	1,877	2,027	1,318	1,480	1,691	4.8%
Fuel Oil	121	155	173	151	147	2.7%
LPG	464	480	514	487	519	6.9%
Biomass	325	330	332	334	336	0.1%
Ricehull	5	5	5	5	4	-4.0%
Charcoal	155	157	158	159	160	-0.1%
Fuelwood	165	167	169	170	172	0.5%
Biodiesel	37	40	26	29	34	5.5%
Electricity	1,782	1,816	2,089	2,256	2,410	4.1%
Agriculture	436	556	379	402	439	2.2%
Oil	211	243	139	145	148	-1.5%
Gasoline	10	15	17	10	4	-0.5%
Kerosene ^(c)	1	0.5	0.1	-	0	-11.4%
Diesel	198	226	121	133	143	-1.5%
Fuel Oil	2	2	1	1	1	-6.8%
Biodiesel	4	4	2	3	3	-2.0%
Electricity	221	308	237	254	288	4.9%
Non-Energy Use	1,372	1,642	1,289	1,397	1,528	9.7%
Oil	1,160	1,430	1,198	1,317	1,445	12.4%
Coal	212	213	91	80	83	-6.0%
Total	32,601	35,028	35,859	37,019	38,898	3.2%

* does not include energy for power application

**average annual growth rate

*** only 1 entry

a) AAGR from 2014 to 2021

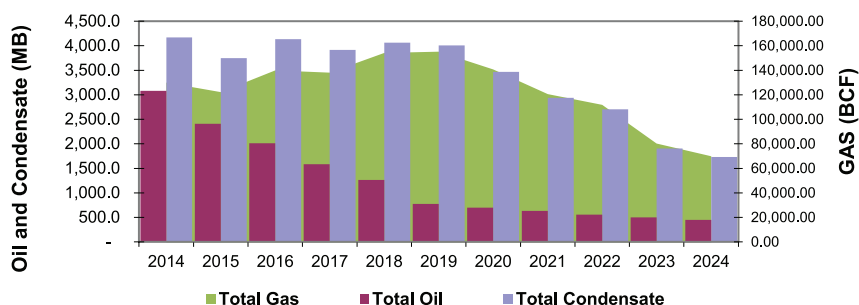
b) AAGR from 2014 to 2022

c) AAGR from 2014 to 2022



OIL AND GAS

Oil and Gas Production, by Source



	2014	2015	2016	2017	2018	2019
In MB						
Total Oil	3,078.68	2,409.75	2,013.56	1,586.61	1,263.53	776.09
Nido ^(a)	79.11	71.15	53.65	56.16	51.74	20.63
Matinloc ^(b)	69.83	70.72	72.75	67.49	43.05	1.54
North Matinloc ^(c)	8.87	8.36	9.12	2.10	-	-
Galoc	2,920.88	2,259.52	1,878.04	1,460.85	1,166.76	744.45
Alegria ^(d)	-	-	-	-	1.98	9.47
Total Condensate	3,078.68	2,409.75	2,013.56	1,586.61	1,263.53	776.09
Galoc	2,920.88	2,259.52	1,878.04	1,460.85	1,166.76	744.45
in MMSCF						
Total Gas	130,351	122,541	140,398	138,497	154,622	155,690
Libertad ^(e)	35	15	-	-	-	-
Malampaya Gas	130,316	122,527	140,398	138,497	154,622	155,690

	2020	2021	2022	2023	2024	AAGR*
In MB						
Total Oil	700.12	632.29	558.28	501.20	447.86	-17.5%
Nido ^(a)	-	-	-	-	-	-23.6%
Matinloc ^(b)	-	-	-	-	-	-53.4%
North Matinloc ^(c)	-	-	-	-	-	-38.1%
Galoc	695.25	630.25	556.90	501.14	447.86	-17.1%
Alegria ^(d)	4.87	2.04	1.37	0.06	-	-49.7%
Total Condensate	3,469.45	2,936.44	2,706.11	1,904.42	1,733.44	-8.4%
Galoc	3,469.45	2,936.44	2,706.11	1,904.42	1,733.44	-8.4%
in MMSCF						
Total Gas	141,191	121,089	112,172	80,660	70,338	-6.7%
Libertad ^(e)	-	-	-	-	-	-58.2%
Malampaya Gas	141,191	121,089	112,172	80,660	70,338	-6.0%

*average annual growth rate

(a) average annual growth rate from 2014 to 2019

(b) average annual growth rate from 2014 to 2019

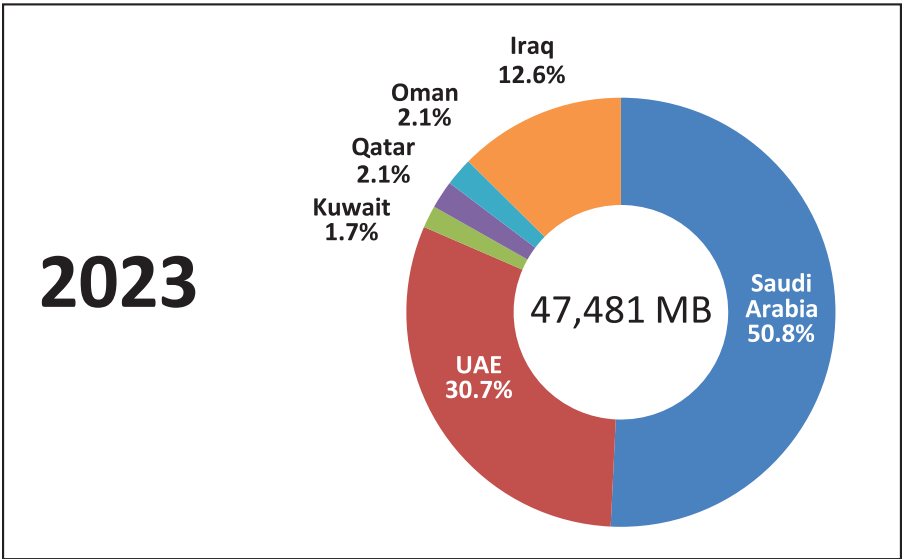
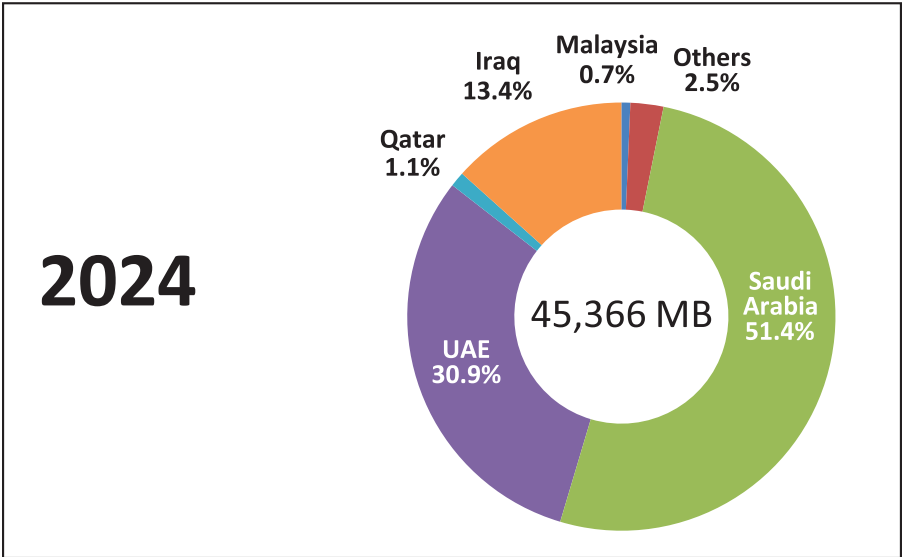
(c) average annual growth rate from 2014 to 2017

(d) average annual growth rate from 2018 to 2023

(e) average annual growth rate from 2014 to 2015



Crude Oil Importation, by Country of Source



in MB

Source	2014	2015	2016	2017	2018	2019
Middle East	49,086	67,855	68,537	69,345	74,555	41,521
Saudi Arabia	37,103	34,427	28,438	27,097	28,880	15,498
Kuwait ^(b)	-	16,877	26,448	24,475	22,589	15,925
UAE	6,403	9,087	10,507	13,549	17,759	9,136
Qatar	5,579	7,464	2,618	2,999	4,235	-
Oman ^(c)	-	-	524	1,225	1,091	961
Indonesia ^(d)	-	-	396	-	221	-
Malaysia ^(e)	3,583	5,025	4,160	916	3,215	4,085
Others**	12,194	5,031	5,544	7,255	7,669	15,061
Total	64,862	77,911	78,637	77,516	85,660	60,666

Source	2020	2021	2022	2023	2024	AAGR*
Middle East	24,040	28,411	43,343	47,481	43,931	-1.1%
Saudi Arabia	15,044	19,059	22,760	24,102	23,340	-4.5%
Iraq ^(a)	-	630	4,760	5,976	6,060	112.7%
Kuwait ^(b)	7,991	-	1,952	798	-	-31.7%
UAE	506	4,769	13,368	14,597	14,020	8.2%
Qatar	498	952	502	999	510	-21.3%
Oman ^(c)	-	3,002	-	1,009	-	9.8%
Indonesia ^(d)	-	-	-	-	-	-25.3%
Malaysia ^(e)	629	-	-	-	306	-25.2%
Others**	8,273	1,277	-	-	1,130	-21.2%
Total	32,942	29,689	43,343	47,481	45,366	-3.5%

*average annual growth rate

(a) AAGR from 2021 to 2024

(b) AAGR from 2015 to 2023

(c) AAGR from 2016 to 2023

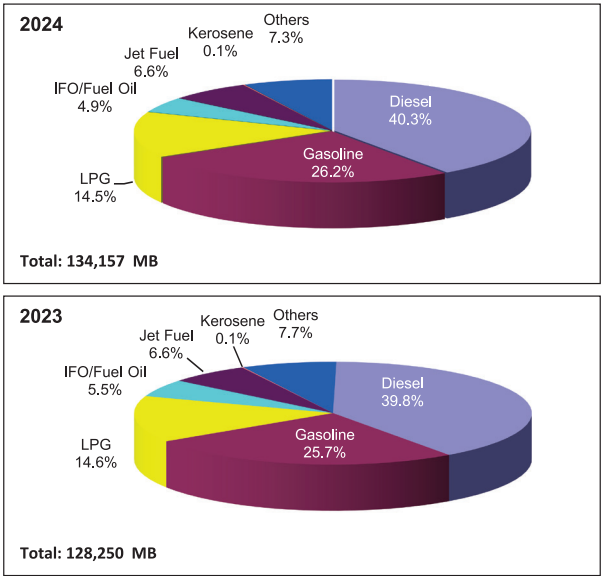
(d) AAGR from 2016 to 2018

(e) AAGR from 2014 to 2020

** includes Singapore, Brunei, Russia, Vietnam, Korea, Australia and other Asia and Pacific Region



Petroleum Products Importation, by Fuel Type



in MB

Fuel	2014	2015	2016	2017	2018	2019
Diesel	30,343	28,375	35,345	40,105	38,784	49,462
Gasoline	14,828	15,148	15,705	17,162	19,004	18,726
LPG	9,299	9,691	11,613	13,910	15,224	15,957
IFO/Fuel Oil	4,901	10,129	7,162	6,921	5,244	5,061
Jet Fuel	6,579	5,722	6,837	8,928	9,331	11,708
Kerosene	430	199	252	317	265	134
Others**	3,279	8,670	9,194	10,073	9,721	7,889
Total	69,658	77,934	86,108	97,415	97,573	108,936

Fuel	2020	2021	2022	2023	2024	AAGR*
Diesel	41,977	50,655	50,976	51,071	54,060	5.9%
Gasoline	25,038	24,927	31,002	32,925	35,120	9.0%
LPG	17,109	19,723	18,797	18,722	19,495	7.7%
IFO/Fuel Oil	6,481	7,097	7,128	7,039	6,567	3.0%
Jet Fuel	3,408	3,331	7,128	8,405	8,917	3.1%
Kerosene	304	398	197	167	162	-9.3%
Others**	8,791	11,442	7,965	9,922	9,835	11.6%
Total	103,108	117,572	123,192	128,250	134,157	6.8%

*average annual growth rate
**others include asphalt, solvents, naptha/reformate, condensate

Petroleum Products Importation, by Country of Source

MB

Source	2014	2015	2016	2017	2018	2019
Middle East	1,966	1,678	6,280	5,272	11,521	8,436
Bahrain	-	-	-	-	-	-
Iran	-	-	1	3	1	5
Iraq	-	-	-	-	163	-
KSA	551	1,085	1,402	1,081	1,204	1,242
Kuwait	406	228	692	1,150	443	1,323
Oman	-	135	-	90	3	176
Qatar	450	-	1,490	1,263	1,265	657
UAE	559	231	2,695	1,684	8,442	5,033
ASEAN	16,523	19,979	19,556	19,732	18,321	24,697
Brunei	-	-	-	-	-	222
Indonesia	697	228	1,610	1,186	539	590
Malaysia	1,758	7,523	4,572	7,557	7,661	9,161
Philippines	805	-	-	-	-	-
Singapore	9,982	10,323	12,147	10,273	7,910	12,551
Thailand	2,473	1,272	36	377	930	266
Vietnam	808	633	1,192	340	1,281	1,907
OTHER ASIA	50,335	53,160	58,342	67,305	64,491	73,156
China	10,504	10,938	24,997	32,460	29,421	45,472
Hong Kong	1	-	51	158	-	0
India	1,064	3,258	2,763	4,686	2,682	3,219
Japan	368	1,824	4,701	4,261	2,673	1,373
Russia	-	605	-	-	-	-
South Korea	21,229	17,886	16,233	22,521	28,083	22,701
Sri Lanka	-	-	-	-	271	-
Taiwan	16,993	17,674	9,204	2,216	1,028	391
Pakistan	177	975	393	1,002	333	-
OTHERS**	834	3,117	1,930	5,106	3,240	2,646
Total	69,658	77,934	86,108	97,415	97,573	108,936



Source	2020	2021	2022	2023	2024	AAGR*
Middle East	8,492	7,695	3,727	5,287	5,864	11.5%
Bahrain	-	4	-	5	992	
Iraq	-	-	-	44	-	
KSA	1,407	2,438	116	1,092	786	
Kuwait	1,384	1,001	490	1,764	975	
Oman	-	-	-	-	240	
Qatar	1,868	1,202	2,293	1,790	2,364	
UAE	3,833	3,049	828	593	506	
ASEAN	34,651	43,833	44,630	50,053	41,201	9.6%
Brunei	4,293	5,660	4,245	3,408	3,176	
Indonesia	1,738	-	-	141	26	
Malaysia	9,338	13,429	11,723	16,049	16,611	
Philippines	-	-	-	-	-	
Singapore	18,188	20,361	27,437	27,323	19,620	
Thailand	409	3,647	1,110	2,521	1,769	
Vietnam	685	736	116	612	-	
OTHER ASIA	55,327	61,344	72,837	71,140	86,402	5.6%
China	32,247	35,279	18,974	31,470	32,945	
Hong Kong	20	0	-	-	-	
India	4,409	4,950	1,009	1,950	4,365	
Japan	518	2,323	7,722	4,137	3,920	
Russia	512	-	1,320	241	-	
South Korea	16,045	15,267	37,835	29,780	38,754	
Taiwan	1,575	3,525	5,978	3,560	6,418	
OTHERS**	4,638	4,701	1,998	1,770	690	-1.9%
Total	103,108	117,572	123,192	128,250	134,157	6.8%

*average annual growth rate

**Others include countries from Africa, Asia and Pacific, Europe and North America



Petroleum Products Exportation, by Country of Destination

MB

Destination	2014	2015	2016	2017	2018	2019
MIDDLE EAST ^(a)	-	-	-	122	-	-
UAE	-	-	-	122	-	-
ASEAN	5,916	6,691	6,027	7,525	9,552	8,572
Indonesia	21	98	67	119	252	19
Malaysia	679	2,416	1,259	2,131	1,319	1,786
Singapore	3,704	3,066	2,711	2,849	4,812	4,602
Thailand	1,513	919	1,683	2,333	3,025	1,818
Vietnam	-	192	308	93	144	347
OTHER ASIA	3,643	6,433	7,727	6,899	7,040	3,037
China	717	1,441	1,897	2,670	3,986	1,802
Hong Kong	-	-	-	-	-	-
Japan	-	-	20	62	60	-
South Korea	2,284	3,453	3,385	2,645	1,911	907
Taiwan	643	1,539	2,424	1,523	1,083	327
OTHERS ^(b)	2	864	18	84	160	67
Total	9,561	13,988	13,772	14,631	16,752	11,676

Destination	2020	2021	2022	2023	2024	AAGR*
MIDDLE EAST	-	298	-	11	-	-33.0%
UAE	-	-	-	11	-	
Iraq	-	298	-	-	-	
ASEAN	5,911	3,729	3,698	2,459	3,025	-6.5%
Brunei	-	324	1,267	291	913	
Indonesia	-	95	90	96	206	
Malaysia	2,571	627	712	93	649	
Singapore	2,753	640	253	1,171	577	
Thailand	476	1,902	1,262	809	651	
Vietnam	111	141	113	-	29	
OTHER ASIA	1,994	2,876	1,977	2,828	3,210	-1%
China	903	1,259	1,287	1,492	2,203	
Hong Kong	306	-	-	-	-	
India	36	146	183	231	554	
Japan	130	-	11	32	-	
South Korea	21	789	246	966	277	
Taiwan	598	682	250	107	176	
OTHERS ^(b)	0	77	-	472	-	-7.3%
Total	7,905	6,980	5,675	5,770	6,236	-4.2%

*average annual growth rate

(a) AAGR from 2017 to 2023

(b) AAGR from 2015 to 2023

Others include Australia, Belgium, Guam, Egypt, Saipan and USA



Petroleum Products Consumption, by Sector and Fuel Type

MB

	2014	2015	2016	2017	2018	2019
Industry	9,224	10,528	11,046	11,022	11,392	10,732
Kerosene	126	139	148	170	16	15
LPG	1,387	1,528	1,412	924	2,129	1,872
Diesel	3,975	4,357	4,869	5,570	5,144	5,598
Fuel Oil	3,649	4,413	4,520	4,247	4,000	3,136
Biodiesel	86	92	97	111	103	112
Transport	79,996	91,891	99,456	105,255	108,811	112,365
Gasoline	25,795	29,601	32,568	35,411	36,516	39,504
Diesel	36,097	43,435	45,749	46,067	48,043	48,205
Fuel Oil	1,690	1,386	1,707	2,431	1,845	1,637
Aviation Fuel	12,463	13,086	14,879	16,474	17,390	17,674
LPG	453	321	146	112	108	95
Bioethanol	2,765	3,168	3,477	3,818	3,936	4,276
Biodiesel	734	893	929	942	972	974
Households	9,074	10,301	11,938	12,342	13,373	14,014
LPG	8,343	9,632	11,314	11,749	12,754	13,445
Kerosene	731	669	624	593	620	569
Services	11,641	10,692	13,456	17,357	18,393	19,386
LPG	2,890	3,360	4,054	5,767	5,495	5,370
Diesel	7,819	6,262	7,853	10,368	11,641	12,668
Fuel Oil	780	959	1,390	1,011	1,025	1,095
Biodiesel	151	111	158	212	233	253
Agriculture	1,306	1,471	1,731	2,202	1,579	1,741
Gasoline	37	66	61	98	58	74
Kerosene	3	3	5	3	3	3
Diesel	1,227	1,355	1,577	2,008	1,484	1,615
Fuel Oil	13	18	56	52	5	17
Biodiesel	26	29	32	41	30	32
Power Generation	9,762	9,976	8,833	6,965	5,292	5,728
Diesel	2,477	2,137	3,573	2,926	2,770	3,015
Fuel Oil	7,233	7,793	5,188	3,979	2,460	2,646
Biodiesel	52	45	72	60	62	67
Non-Energy Use	3,501	8,368	8,954	11,397	9,964	7,851
Total	124,503	143,226	155,414	166,539	168,805	171,817



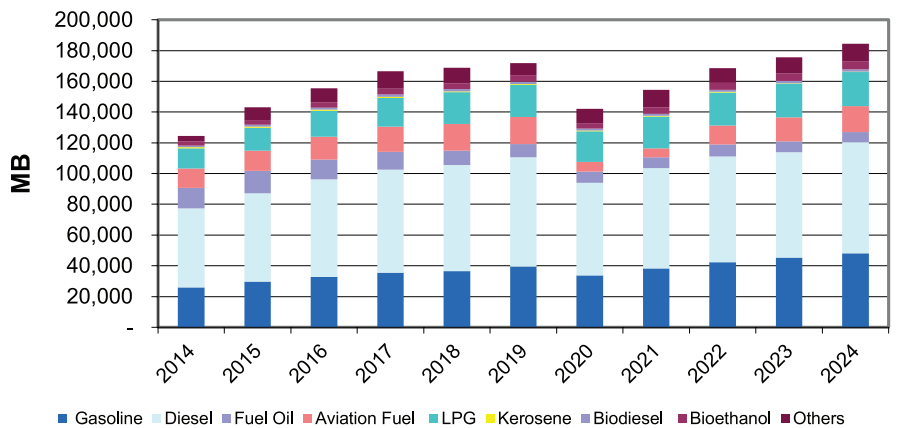
MB

	2020	2021	2022	2023	2024	AAGR*
Industry	11,872	12,149	14,216	12,548	11,723	2.4%
Kerosene	131	157	87	120	96	-2.7%
LPG	1,934	2,165	2,195	2,154	2,034	3.9%
Diesel	6,163	6,593	8,906	8,056	7,396	6.4%
Fuel Oil	3,543	3,115	2,845	2,053	2,046	-5.6%
Biodiesel	101	118	182	164	150	5.7%
Transport	82,310	90,845	106,848	114,299	120,771	4.2%
Gasoline	33,609	38,224	42,096	45,362	48,062	6.4%
Diesel	36,625	40,363	44,656	45,601	47,958	2.9%
Fuel Oil	1,452	1,565	2,426	1,667	1,709	0.1%
Aviation Fuel	6,188	5,741	12,199	15,824	16,871	3.1%
LPG	48	30	2	0	0	-100.0%
Bioethanol	3,643	4,147	4,569	4,919	5,197	6.5%
Biodiesel	745	775	899	926	974	2.9%
Households	13,272	13,583	14,084	14,542	14,805	5.0%
LPG	12,870	13,178	13,723	14,291	14,570	5.7%
Kerosene	402	405	361	251	235	-10.7%
Services	20,087	21,638	16,756	17,534	19,464	5.3%
LPG	5,028	5,208	5,574	5,280	5,633	6.9%
Diesel	13,934	15,052	9,782	10,985	12,553	4.8%
Fuel Oil	840	1,071	1,200	1,044	1,019	2.7%
Biodiesel	284	307	200	224	258	5.5%
Agriculture	1,602	1,851	1,065	1,101	1,123	-1.5%
Gasoline	80	126	143	86	36	-0.5%
Kerosene	4	4	1	0	0	-33.5%
Diesel	1,473	1,676	895	991	1,060	-1.5%
Fuel Oil	15	12	7	5	6	-6.8%
Biodiesel	30	34	18	20	21	-2.0%
Power Generation	3,646	2,969	6,106	5,079	5,158	-6.2%
Diesel	2,149	1,610	4,579	2,865	3,269	2.8%
Fuel Oil	1,453	1,282	1,434	2,155	1,824	-12.9%
Biodiesel	44	76	93	58	66	2.3%
Non-Energy Use	9,228	11,387	9,496	10,477	11,471	12.6%
Total	142,017	154,422	168,571	175,580	184,515	4.0%

*average annual growth rate



Petroleum Products Consumption, by Fuel Type



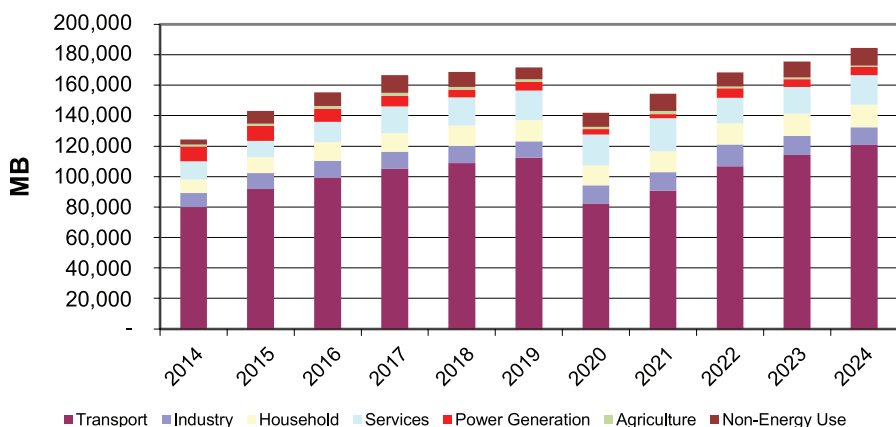
	2014	2015	2016	2017	2018	2019
Gasoline	25,833	29,667	32,630	35,509	36,574	39,578
Diesel	51,595	57,545	63,622	66,939	69,082	71,101
Fuel Oil	13,364	14,568	12,862	11,719	9,335	8,530
Aviation Fuel	12,463	13,086	14,879	16,474	17,390	17,674
LPG	13,073	14,842	16,926	18,552	20,486	20,782
Kerosene	860	811	777	767	638	587
Biodiesel	1,049	1,171	1,289	1,364	1,400	1,437
Bioethanol	2,765	3,168	3,477	3,818	3,936	4,276
Others**	3,501	8,368	8,954	11,397	9,964	7,851
Total	124,503	143,226	155,414	166,539	168,805	171,817

	2020	2021	2022	2023	2024	AAGR*
Gasoline	33,688	38,349	42,238	45,448	48,098	6.4%
Diesel	60,345	65,294	68,819	68,497	72,236	3.4%
Fuel Oil	7,303	7,047	7,912	6,924	6,605	-6.8%
Aviation Fuel	6,188	5,741	12,199	15,824	16,871	3.1%
LPG	19,881	20,582	21,495	21,726	22,237	5.5%
Kerosene	537	565	450	371	332	-9.1%
Biodiesel	1,204	1,310	1,392	1,393	1,468	3.4%
Bioethanol	3,643	4,147	4,569	4,919	5,197	6.5%
Others**	9,228	11,387	9,496	10,477	11,471	12.6%
Total	142,017	154,422	168,571	175,580	184,515	4.0%

*average annual growth rate
 **includes asphalts, solvents, naphtha/reformate, condensate



Petroleum Products Consumption, by Sector



MB

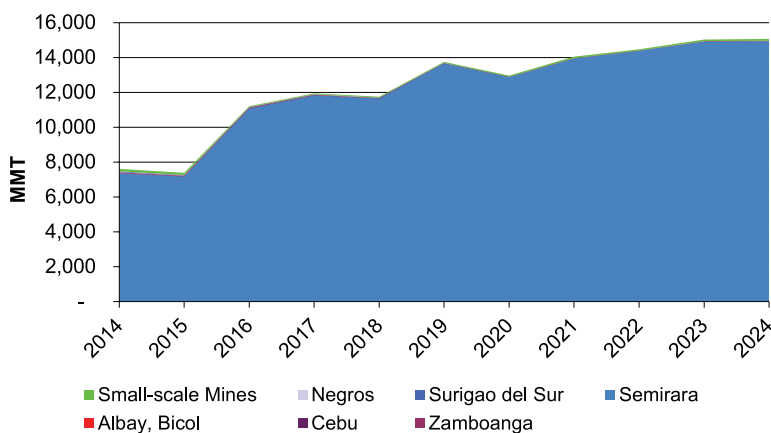
	2014	2015	2016	2017	2018	2019
Transport	79,996	91,891	99,456	105,255	108,811	112,365
Industry	9,224	10,528	11,046	11,022	11,392	10,732
Households	9,074	10,301	11,938	12,342	13,373	14,014
Services	11,641	10,692	13,456	17,357	18,393	19,386
Agriculture	1,306	1,471	1,731	2,202	1,579	1,741
Power Generation	9,762	9,976	8,833	6,965	5,292	5,728
Non-Energy Use	3,501	8,368	8,954	11,397	9,964	7,851
Total	124,503	143,226	155,414	166,539	168,805	171,817

	2020	2021	2022	2023	2024	AAGR*
Transport	82,310	90,845	106,848	114,299	120,771	4.2%
Industry	11,872	12,149	14,216	12,548	11,723	2.4%
Households	13,272	13,583	14,084	14,542	14,805	5.0%
Services	20,087	21,638	16,756	17,534	19,464	5.3%
Agriculture	1,602	1,851	1,065	1,101	1,123	-1.5%
Power Generation	3,646	2,969	6,106	5,079	5,158	-6.2%
Non-Energy Use	9,228	11,387	9,496	10,477	11,471	12.6%
Total	142,017	154,422	168,571	175,580	184,515	4.0%

*average annual growth rate



Coal Production, by Source



in MMT at 10,000 BTU/lb

	2014	2015	2016	2017	2018	2019
Semirara	7,345	7,168	11,084	11,839	11,654	13,670
Zamboanga	15	5	0	-	-	-
Cebu	44	29	35	13	7	6
Albay, Bicol	21	28	16	12	13	9
Surigao del Sur ^(a)	50	28	21	23	26	-
Negros ^(b)	-	-	-	0.49	0.14	0.68
Small-scale Mines	127	119	54	44	55	65
Total Production	7,601	7,378	11,211	11,932	11,755	13,751
Run of Mine (MMT)	8,419	8,173	12,087	13,264	13,054	15,274

	2020	2021	2022	2023	2024	AAGR*
Semirara	12,880	13,957	14,387	14,918	14,918	7.3%
Zamboanga	-	-	-	0.99	18.63	2.0%
Cebu	2	1	2	4	4	-20.8%
Albay, Bicol	13	8	2	10	6	-11.4%
Surigao del Sur ^(a)	-	-	-	-	-	-15.0%
Negros ^(b)	0	0	0	1	2	24.1%
South Cotabato ^(c)	-	-	-	229	851	272.3%
Small-scale Mines	57	82	66	93	117	-0.8%
Total Production	12,951	14,048	14,457	15,255	15,918	7.7%
Run of Mine (MMT)	13,257	14,378	16,061	16,509	17,661	7.7%

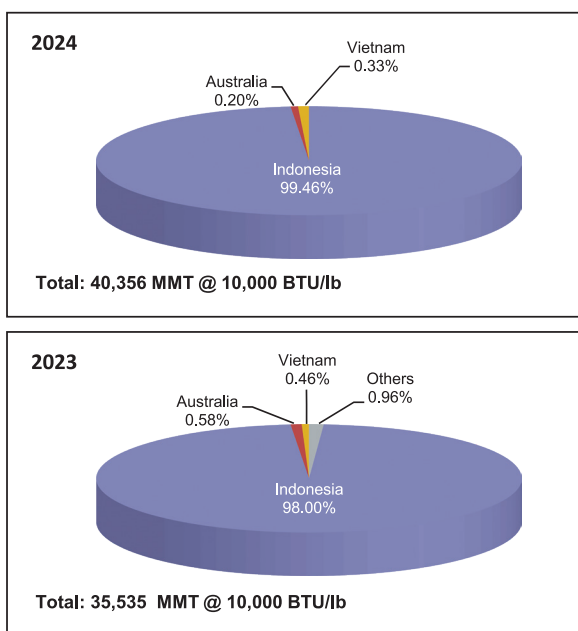
*average annual growth rate

(a) AAGR from 2014 to 2018

(b) AAGR from 2017 to 2024

(c) AAGR from 2023 to 2024

Coal Importation, by Country of Source



in MMT at 10,000 BTU/lb

Country	2014	2015	2016	2017	2018	2019
Indonesia	14,975	16,673	17,988	19,663	23,285	26,305
Australia ^(a)	-	306	1,310	1,401	1,249	711
China ^(b)	-	-	-	-	96	1
Vietnam	191.41	168.07	270.19	219.38	303.26	181.70
Others**	15	132	462	984	1,368	494
Total	15,182	17,279	20,030	22,268	26,301	27,692

Country	2020	2021	2022	2023	2024	AAGR*
Indonesia	28,604	30,514	31,955	34,822	40,139	10.4%
Australia ^(a)	538	242	433	207	82	-13.7%
China ^(b)	-	0.05	0.11	0.35	0.33	-61.2%
Vietnam	103	297	106	164	134	-3.5%
Others**	279	186	202	341	1	-21.4%
Total	29,524	31,239	32,696	35,535	40,356	10.3%

*average annual growth rate

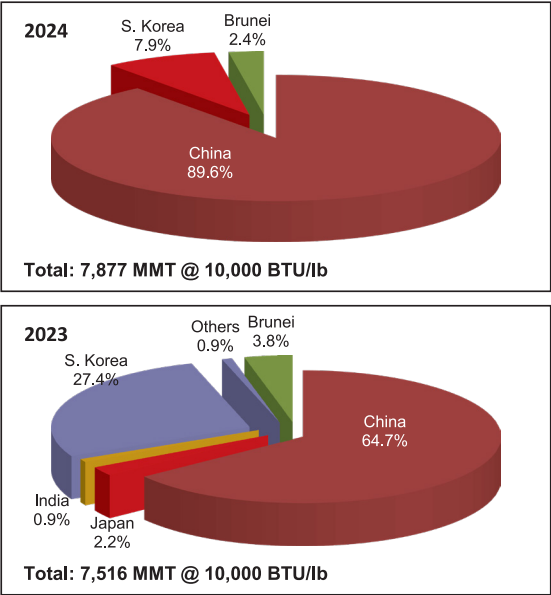
(a) AAGR from 2015 to 2024

(b) AAGR from 2018 to 2024

**Imports from India, Malaysia, Peru, Russia, Taiwan, South Korea, South Africa and USA



Coal Exportation, by Country of Destination



in MMT at 10,000 BTU/lb

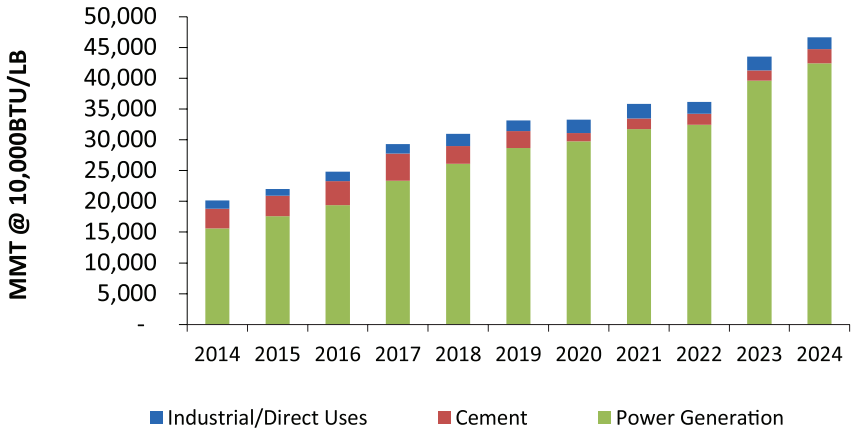
Country	2014	2015	2016	2017	2018	2019
China	4,318	2,574	6,121	5,333	4,610	9,012
India ^(a)	51	-	44	148	51	51
Hongkong ^(b)	252	-	-	-	-	-
Thailand ^(c)	288	300	208	98	51	213
Taiwan ^(d)	73	-	-	177	-	65
S. Korea	193	-	-	-	-	-
Others*** ^(e)	203	22	-	-	-	-
Total	5,377	2,896	6,373	5,755	4,713	9,341

Country	2020	2021	2022	2023	2024	AAGR*
Brunei ^(f)	-	-	96	286	193	41.7%
China	6,784	8,246	3,706	4,866	7,058	5.0%
India ^(a)	47	-	69	71	-	3.7%
Thailand ^(c)	109	-	452	-	-	5.8%
Taiwan ^(d)	-	-	-	-	-	-2.3%
Japan ^(g)	-	-	-	163	-	-
S. Korea	71	355	2,026	2,061	626	12.5%
Others*** ^(f)	34	103	291	69	-	-11.3%
Total	7,044	8,703	6,640	7,516	7,877	3.9%

*average annual growth rate (AAGR)
 (a) AAGR from 2014 to 2023, (b) only one (1) entry (2014), (c) AAGR from 2014 to 2022,
 (d) AAGR from 2014 to 2019, (e) AAGR from 2014 to 2023, (f) AAGR from 2022 to 2024
 (g) only one (1) entry (2023)
 **includes Cambodia, Papua New Guinea, and Vietnam



Coal Consumption, by Major Type of User



in MMT @ 10,000 BTU/lb

	2014	2015	2016	2017	2018	2019
Power Generation	15,587	17,554	19,386	23,327	26,101	28,654
Cement	3,203	3,348	3,893	4,423	2,848	2,754
Industrial/Direct Uses*	1,372	1,104	1,515	1,569	2,027	1,714
TOTAL	20,163	22,006	24,794	29,320	30,976	33,122

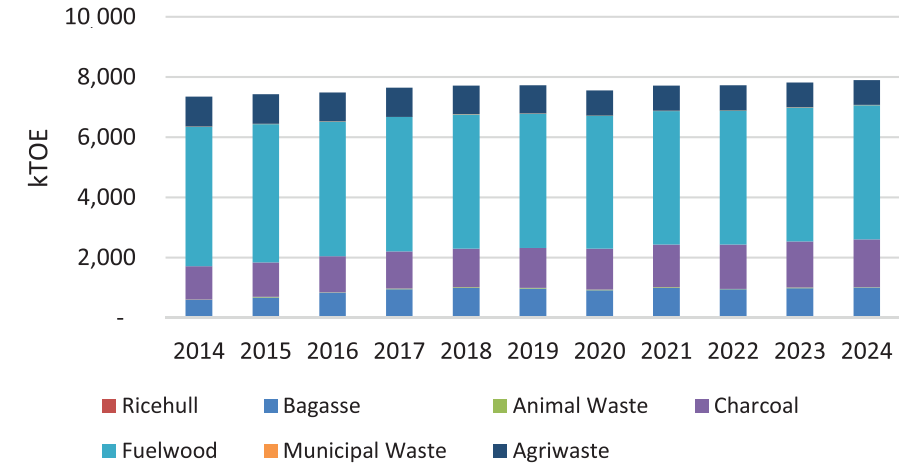
	2020	2021	2022	2023	2024	AAGR**
Power Generation	29,755	31,738	32,454	39,573	42,446	10.5%
Cement	1,312	1,697	1,755	1,716	2,295	-3.3%
Industrial/Direct Uses*	2,177	2,400	1,932	2,200	1,930	3.5%
TOTAL	33,244	35,834	36,141	43,490	46,670	8.8%

*includes non-energy use as raw material

**average annual growth rate



Biomass Production, by Fuel Type



	2014	2015	2016	2017	2018	2019
Fuelwood	4,623	4,596	4,472	4,465	4,458	4,454
Charcoal	1,107	1,138	1,193	1,236	1,281	1,329
Agriwaste	992	982	967	968	958	947
Bagasse	540	627	780	899	942	919
Ricehull	55	55	56	57	58	58
Animal Waste	20	21	21	21	21	22
Municipal Waste	19	11	5	6	6	7
Total	7,356	7,431	7,494	7,651	7,725	7,736

	2020	2021	2022	2023	2024	AAGR*
Fuelwood	4,421	4,439	4,449	4,450	4,459	-0.4%
Charcoal	1,364	1,420	1,476	1,531	1,592	3.7%
Agriwaste	836	837	831	824	819	-1.9%
Bagasse	877	959	908	947	961	5.9%
Ricehull	43	44	44	44	44	-2.1%
Animal Waste	16	17	17	17	17	-1.9%
Municipal Waste	5	4	7	5	7	-9.8%
Total	7,563	7,721	7,731	7,819	7,899	0.7%

*average annual growth rate (AAGR)

Geothermal

	2015	2016	2017	2018	2019
Installed Generating Capacity (MW)	1,917	1,916	1,916	1,944	1,928
Dependable Generating Capacity (MW)	1,601	1,689	1,752	1,770	1,792
Electricity Generation (GWh)	11,044	11,070	10,270	10,435	10,691

	2020	2021	2022	2023	2024
Installed Generating Capacity (MW)	1,928	1,928	1,952	1,952	1,952
Dependable Generating Capacity (MW)	1,753	1,753	1,763	1,708	1,708
Electricity Generation (GWh)	10,757	10,016	10,425	10,730	10,789

Hydropower

	2015	2016	2017	2018	2019
Installed Generating Capacity (MW)	3,600	3,618	3,627	3,701	3,760
Dependable Generating Capacity (MW)	3,073	3,181	3,269	3,473	3,508
Electricity Generation (GWh)	8,665	8,111	9,611	9,384	8,025

	2020	2021	2022	2023	2024
Installed Generating Capacity (MW)	3,779	3,752	3,745	3,799	3,836
Dependable Generating Capacity (MW)	3,527	3,500	3,444	3,499	3,485
Electricity Generation (GWh)	7,192	9,185	10,085	10,287	10,909

Wind

	2015	2016	2017	2018	2019
Installed Generating Capacity (MW)	427	427	427	427	427
Dependable Generating Capacity (MW)	379	383	383	427	427
Electricity Generation (GWh)	748	975	1,094	1,153	1,042

	2020	2021	2022	2023	2024
Installed Generating Capacity (MW)	443	427	427	427	427
Dependable Generating Capacity (MW)	443	427	412	412	412
Electricity Generation (GWh)	1,026	1,270	1,030	1,308	1,239



Solar

	2015	2016	2017	2018	2019
Installed Generating Capacity (MW)	165	765	885	896	921
Dependable Generating Capacity (MW)	125	594	700	740	737
Electricity Generation (GWh)	139	1,097	1,201	1,249	1,246

	2020	2021	2022	2023	2024
Installed Generating Capacity (MW)	1,019	1,317	1,530	1,653	2,710
Dependable Generating Capacity (MW)	817	1,034	1,150	1,249	2,154
Electricity Generation (GWh)	1,373	1,470	1,822	2,544	3,811

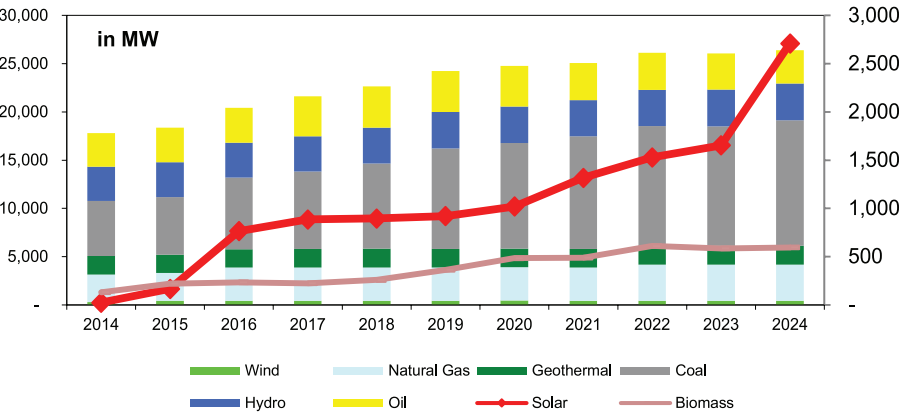
Biomass

	2015	2016	2017	2018	2019
Installed Generating Capacity (MW)	221	233	224	258	363
Dependable Generating Capacity (MW)	146	157	160	182	227
Electricity Generation (GWh)	367	726	1,013	1,105	1,040

	2020	2021	2022	2023	2024
Installed Generating Capacity (MW)	483	489	611	585	595
Dependable Generating Capacity (MW)	285	291	382	374	378
Electricity Generation (GWh)	1,261	1,445	1,322	1,409	1,446



Installed Generating Capacity, by Source



	2014	2015	2016	2017	2018	2019
Total Installed Capacity	17,944	18,765	21,423	22,728	23,815	25,531
Coal	5,708	5,963	7,419	8,049	8,844	10,417
Oil	3,476	3,610	3,616	4,153	4,292	4,262
Natural Gas	2,862	2,862	3,431	3,447	3,453	3,453
Renewable Energy	5,898	6,330	6,958	7,079	7,227	7,399
Geothermal	1,918	1,917	1,916	1,916	1,944	1,928
Hydro	3,543	3,600	3,618	3,627	3,701	3,760
Wind	283	427	427	427	427	427
Solar ^(a)	23	165	765	885	896	921
Biomass	131	221	233	224	258	363

	2020	2021	2022	2023	2024	AAGR*
Total Installed Capacity	26,286	26,882	28,258	28,291	29,706	5%
Coal	10,944	11,669	12,428	12,406	13,006	9%
Oil	4,237	3,847	3,834	3,737	3,448	0%
Natural Gas	3,453	3,453	3,732	3,732	3,732	3%
Renewable Energy	7,653	7,914	8,264	8,417	9,520	5%
Geothermal	1,928	1,928	1,952	1,952	1,952	0%
Hydro	3,779	3,752	3,745	3,799	3,836	1%
Wind	443	427	427	427	427	4%
Solar ^(a)	1,019	1,317	1,530	1,653	2,710	17%
Biomass	484	489	611	585	595	16%

*average annual growth rate
(a) average annual growth rate from 2015 to 2024

Power Generation, by Source and Grid

in GWh

Luzon	2014	2015	2016	2017	2018	2019
Coal	27,346	29,680	33,143	33,953	37,362	40,508
Oil	2,342	1,845	2,562	2,379	2,188	2,674
Natural Gas	18,686	18,878	19,854	20,547	21,334	22,354
Renewable Energy	8,392	9,711	10,938	11,633	11,845	10,640
Geothermal	3,817	4,096	4,227	3,910	3,871	3,647
Hydro	4,357	4,769	5,011	5,730	5,945	5,084
Biomass	65	187	439	599	594	592
Solar ^(a)	-	66	495	496	503	493
Wind	152	592	767	899	931	824
Total	56,766	60,113	66,498	68,512	72,728	76,177

Luzon	2020	2021	2022	2023	2024	AAGR*
Coal	40,576	43,133	48,626	53,072	57,070	8%
Oil	1,804	996	1,507	708	603	-13%
Natural Gas	19,497	19,060	17,884	16,668	18,047	0%
Renewable Energy	10,542	12,053	11,803	12,842	14,550	6%
Geothermal	3,808	3,838	3,921	4,286	4,657	2%
Hydro	4,510	5,412	5,239	5,080	5,308	2.0%
Biomass	780	988	691	684	645	26%
Solar ^(a)	588	721	1,099	1,678	2,905	52%
Wind	855	1,095	854	1,114	1,034	21%
Total	72,419	75,243	79,821	83,290	90,269	5%

in GWh

Visayas	2014	2015	2016	2017	2018	2019
Coal	4,449	4,968	5,270	6,624	6,785	7,962
Oil	766	672	637	541	353	524
Natural Gas ^(a)	4	-	-	-	-	-
Renewable Energy	5,794	6,530	7,047	6,889	7,129	7,573
Geothermal	5,627	6,105	5,974	5,564	5,737	6,278
Hydro	35	38	64	90	73	57
Biomass	117	159	276	414	439	356
Solar	15	71	525	627	658	665
Wind ^(b)	-	157	209	194	222	218
Total	11,014	12,170	12,955	14,054	14,266	16,060

Visayas	2020	2021	2022	2023	2024	AAGR*
Coal	7,696	8,999	8,501	10,401	8,571	7%
Oil	298	468	563	473	519	-4%
Natural Gas ^(a)	-	-	-	-	-	-
Renewable Energy	7,491	6,801	7,159	7,289	7,009	2%
Geothermal	6,205	5,535	5,813	5,740	5,408	-0.4%
Hydro	65	89	39	141	162	16%
Biomass	374	350	517	488	514	16%
Solar	676	652	614	727	720	47%
Wind ^(b)	171	174	176	194	205	3%
Total	15,485	16,268	16,222	18,163	16,099	4%

in GWh

Mindanao	2014	2015	2016	2017	2018	2019
Coal	1,258	2,038	4,890	6,271	7,785	9,420
Oil	2,599	3,369	2,462	867	633	554
Natural Gas	-	-	-	-	-	-
Renewable Energy	5,624	4,723	3,994	4,666	4,352	3,831
Geothermal	864	842	869	797	826	766
Hydro	4,745	3,858	3,036	3,791	3,366	2,885
Biomass	14	21	11	-	72	93
Solar ^(a)	1	2	77	78	88	87
Wind	0	-	-	-	-	-
Total	9,481	10,130	11,345	11,804	12,770	13,805

Mindanao	2020	2021	2022	2023	2024	AAGR*
Coal	9,904	9,920	9,303	10,281	13,718	27%
Oil	372	152	449	123	220	-22%
Natural Gas	-	-	-	-	-	-
Renewable Energy	3,576	4,532	5,721	6,147	6,634	1.7%
Geothermal	744	643	690	704	724	-1.8%
Hydro	2,617	3,684	4,807	5,067	5,438	1.4%
Biomass	107	107	114	237	286	35%
Solar ^(a)	108	97	110	139	186	12%
Wind	-	-	-	-	-	-
Total	13,852	14,604	15,473	16,551	20,572	8%

in GWh

Philippines	2014	2015	2016	2017	2018	2019
Coal	33,054	36,686	43,303	46,847	51,932	57,890
Oil	5,708	5,886	5,661	3,787	3,173	3,752
Natural Gas	18,690	18,878	19,854	20,547	21,334	22,354
Renewable Energy	19,810	20,963	21,979	23,189	23,326	22,044
Geothermal	10,308	11,044	11,070	10,270	10,435	10,691
Hydro	9,137	8,665	8,111	9,611	9,384	8,025
Biomass	196	367	726	1,013	1,105	1,040
Solar ^(a)	17	139	1,097	1,201	1,249	1,246
Wind	152	748	975	1,094	1,153	1,042
Total	77,261	82,413	90,798	94,370	99,765	106,041
**Self-sufficiency level (%)	53	53	51	54	51	47

Philippines	2020	2021	2022	2023	2024	AAGR*
Coal	58,176	62,052	66,430	73,754	79,359	9%
Oil	2,474	1,616	2,519	1,304	1,342	-13%
Natural Gas	19,497	19,060	17,884	16,668	18,047	0%
Renewable Energy	21,609	23,386	24,684	26,278	28,193	4%
Geothermal	10,757	10,016	10,425	10,730	10,789	0.5%
Hydro	7,192	9,185	10,085	10,287	10,909	1.8%
Biomass	1,261	1,445	1,322	1,409	1,446	22%
Solar ^(a)	1,373	1,470	1,822	2,544	3,811	45%
Wind	1,026	1,270	1,030	1,308	1,239	23%
Total	101,756	106,115	111,516	118,004	126,941	5%
**Self-sufficiency level (%)	47	45	43	42	39	

*average annual growth rate

Luzon

(a) AAGR from 2015 to 2024

Visayas

(a) only one (1) entry

(b) AAGR from 2015 to 2024

Mindanao

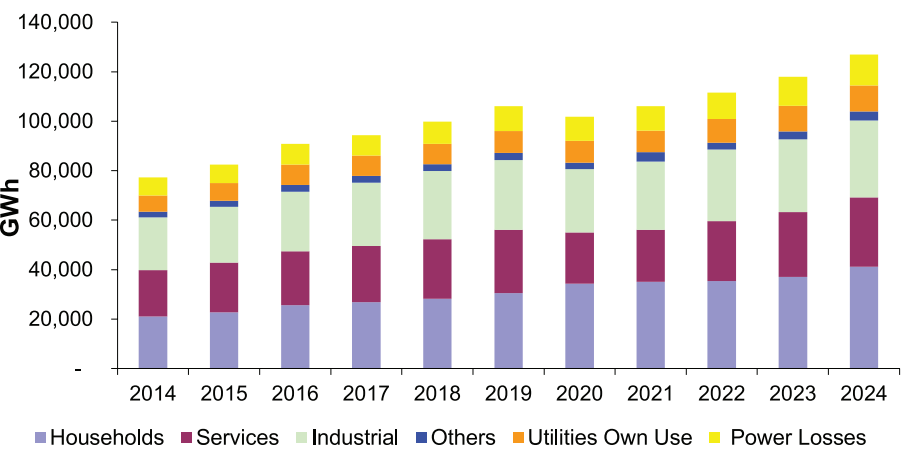
(a) AAGR from 2016 to 2024

Philippines

(a) AAGR from 2015 to 2024



Electricity Consumption, by Sector



in GWh

	2014	2015	2016	2017	2018	2019
Households	20,969	22,747	25,631	26,782	28,261	30,552
Services	18,761	20,085	21,770	22,768	24,016	25,476
Industrial	21,429	22,514	24,117	25,573	27,587	28,194
Others*	2,186	2,462	2,634	2,670	2,753	2,897
Utilities Own Use	6,646	7,124	8,357	8,316	8,141	8,929
Power Losses	7,270	7,481	8,288	8,262	9,007	9,994
Total	77,261	82,413	90,798	94,370	99,765	106,041

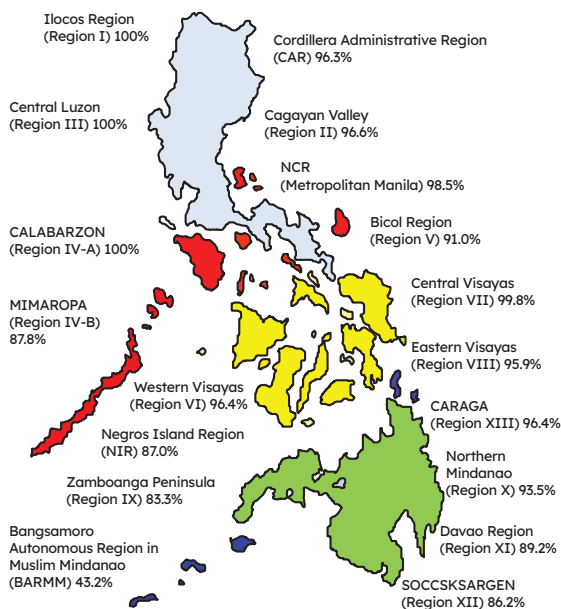
	2020	2021	2022	2023	2024	AAGR**
Households	34,292	34,981	35,324	36,968	41,205	7.0%
Services	20,727	21,119	24,294	26,236	28,033	4.1%
Industrial	25,566	27,623	28,844	29,493	31,074	3.8%
Others*	2,658	3,695	2,871	3,112	3,596	5.1%
Utilities Own Use	8,771	8,729	9,490	10,403	10,558	4.7%
Power Losses	9,742	9,968	10,693	11,793	12,475	5.5%
Total	101,756	106,115	111,516	118,004	126,941	5.1%

* others includeTransport and AFF

**average annual growth rate



Regional Household Electrification Level



Region	"Potential HHs"	"Served HHs "	Unserved HHs (Actual per DU per Province)	Electrification Rate (%)
CAR	504,527	485,827	18,700	96.3%
I	1,441,212	1,450,906	-	100.0%
II	1,043,304	1,007,889	35,415	96.6%
III	3,469,528	3,539,679	-	100.0%
IV-A	4,598,185	4,672,611	-	100.0%
IV-B	887,864	779,948	107,916	87.8%
V	1,492,297	1,358,578	133,719	91.0%
NCR	3,947,966	3,886,962	61,004	98.5%
LUZON	17,384,883	17,182,400	202,483	98.8%
NIR	1,261,845	1,097,256	164,589	87.0%
VI	1,304,302	1,257,561	46,741	96.4%
VII	1,777,144	1,772,742	4,402	99.8%
VIII	1,195,646	1,147,196	48,450	95.9%
VISAYAS	5,538,937	5,274,755	264,182	95.2%
IX	950,310	791,320	158,990	83.3%
X	1,281,438	1,198,767	82,671	93.5%
XI	1,479,195	1,319,870	159,325	89.2%
XII	1,142,888	985,488	157,400	86.2%
CARAGA	798,382	769,643	28,739	96.4%
BARMM	865,385	373,734	491,651	43.2%
MINDANAO	6,517,598	5,438,822	1,078,776	83.4%
PHILIPPINES	29,441,418	27,895,977	1,545,441	94.8%

2024 Notes:

a) Electrification Rate = Served HHs/Potential HHs;

b) Potential HHs is based on PSA Census on Population and Housing projected and adjusted (CPH 2020 for 2024);

c) Unserved HHs = Potential HHs - Served HHs

d) Projections were agreed by NEA and NPC.



Glossary

Condensate	Liquid hydrocarbons separated from gas production.
Dependable Capacity	The capacity that can be relied upon to carry system load for a specified time interval and period, provide assumed reserve, and/or meet firm power obligations.
Electrification	Electrification is either done through grid or off-grid connection. When a barangay is provided with electricity through grid connection, it means that the distribution line has reached the barangay proper. It may also mean that almost 50.0 percent of potential households in the barangay are connected to a distribution utility (DU) (i.e. MERALCO) or at least one is connected to other DUs. Off-grid connection pertains to a barangay having about 20 to 30 households availing the connection.
Energy Intensity	Calculated as units of energy (million tons of oil equivalent, MTOE) per unit of GDP (in billion pesos).
Energy Per Capita	Amount of energy used per person. It is calculated as total primary energy demand (in MTOE) over population (in millions).
Energy Self Sufficiency	The ratio of the country's domestic energy supply to total supply; measures the degree at which domestic energy forms can support total energy demand.
Energy to GDP Elasticity	The percentage change in energy supply to achieve one per cent change in national GDP. Calculated as the ratio of growth of primary energy demand over GDP growth.
Gas (or Natural Gas)	A naturally occurring mixture of hydrocarbon and non-hydrocarbon gases in porous formations beneath the earth's surface, often in association with petroleum. The principal constituent is methane.



Geothermal Energy	Energy generated by heat stored in the earth, or the collection of absorbed heat derived from underground in the atmosphere and oceans.
Gross Domestic Product (GDP)	Total market value of all final goods and services produced within the country in a given period of time (usually a calendar year), or the sum of value added of all final goods and services produced within a country in a given period of time.
Gross National Product (GNP)	The value of all (final) goods and services produced in a country in one year, plus income earned by its citizens abroad, minus income earned by foreigners in the country.
Hydropower	"Also called hydraulic power or water power; derived from the force or energy of moving water, which may be harnessed for useful purposes."
Indigenous Energy	Refers to all energy forms produced/sourced from within the country's natural resources.
Installed Capacity	The total of the capacities shown on the nameplates of the generating units in a powerplant.
Renewable Energy	Energy generated from natural resources which are naturally replenished. It includes solar power, wind power, hydroelectricity, micro hydro, biomass and biofuels.
Run of Mine	Coal directly coming from the mine
Total Final Energy Consumption (TFEC)	The sum of all energy forms consumed/used by different economic sectors
Total Primary Energy Demand (TPED)	The sum of total final consumption, power generation, other energy sector (own use and losses).
Total Primary Energy Supply (TPES)	The sum of all energy derived from domestic sources (indigenous, renewable), imported from outside the country, stock change (+/-) and export (-)



Units of Measurement

BCF Billion Cubic Feet

BTu British Thermal Units

GWh Gigawatt-Hour

KWh Kilowatt-hour

kTOE Thousand tonnes of oil equivalent

Lb Pound

MB Thousand Barrels

MMMT Million Metric Tons

MMSCF Million Standard Cubic Feet

MMT Thousand Metric Tons

MW Megawatt

Php Philippine Peso

ROM Run of Mine

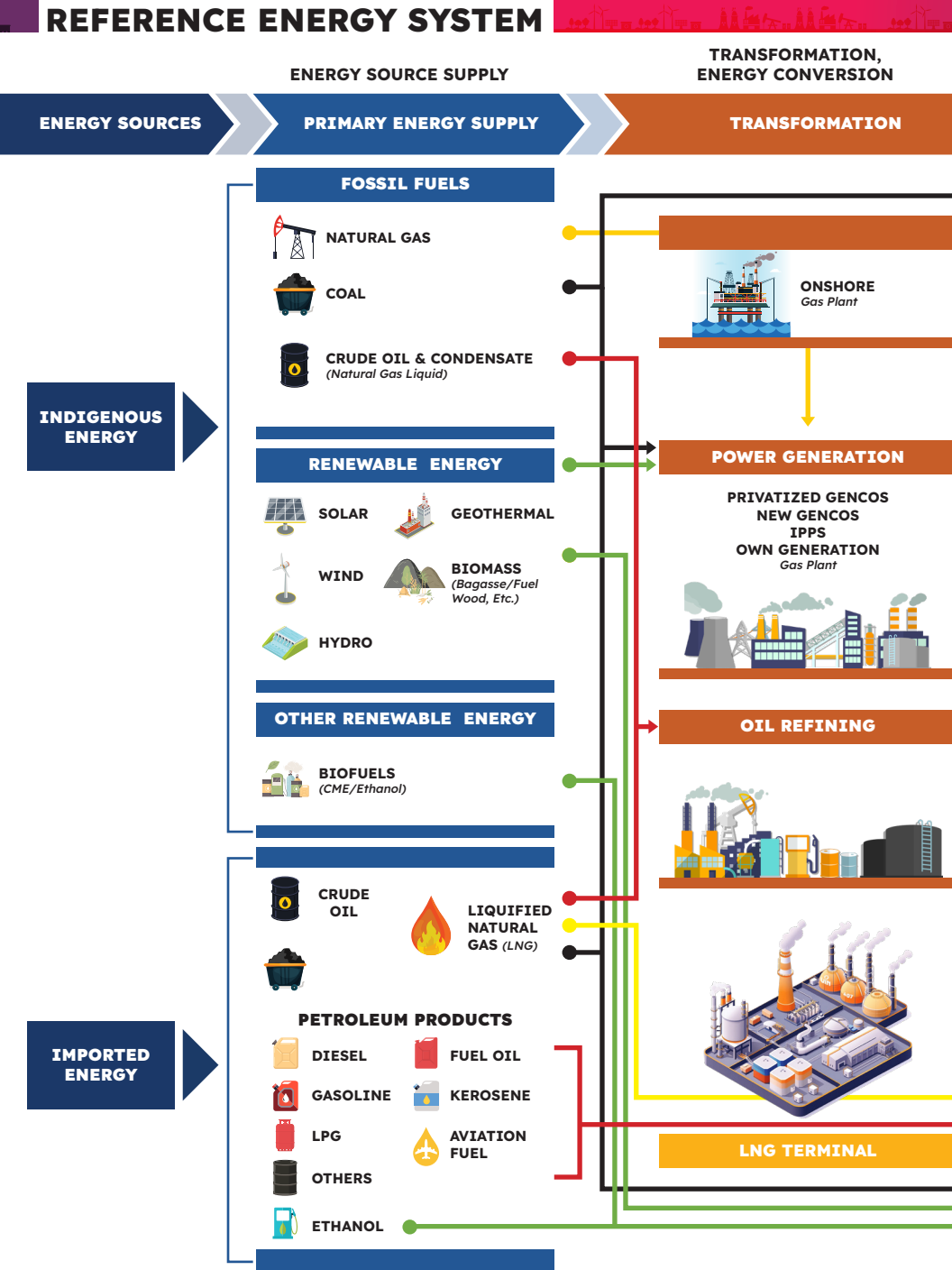
USD US Dollar

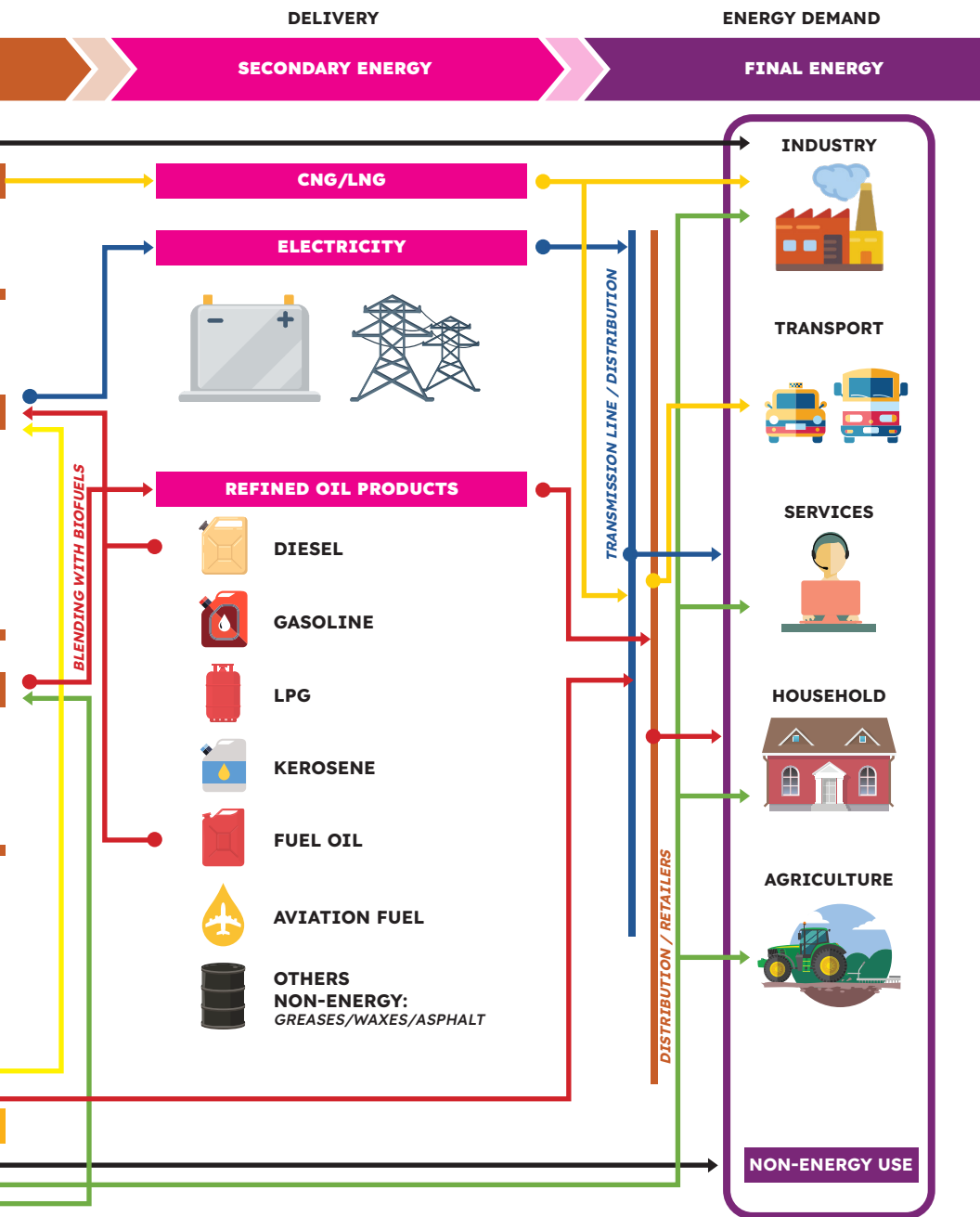


Units of Measurement

Fuels	to kTOE
Coal (MT@10,000 btu/lb.)	0.000528
Natural Gas (MMSCF)	0.023290
Crude (MB)	0.134400
Condensate (NGL) (MB)	0.104400
Premium Gasoline (MB)	0.124500
Regular Gasoline (MB)	0.122300
Kerosene (MB)	0.127000
Diesel (MB)	0.134700
Fuel Oil (MB)	0.144400
LPG (MB)	0.092200
Jet (MB)	0.127000
Avgas (MB)	0.122400
Naphtha (MB)	0.123800
Asphalts (MB)	0.152100
Lubes & Greases (MB)	0.141200
Others (MB)	0.123300
Ricehull (MT)	0.000345
Charcoal (MT)	0.000600
Fuelwood (MT)	0.000329
Bagasse (MT)	0.000426
Agriwaste (MT)	0.000329
Animal Waste (MT)	0.000516
Ethanol (BBL)	0.000089
CME (BBL)	0.000130
Hydro (GWh)	0.086000
Geothermal (GWh)	0.860000
Wind (GWh)	0.860000
Solar (GWh)	0.860000









DEPARTMENT OF ENERGY

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